Civil Systems Engineering
Faculty VI – Planning Building Environment

Impact of Automated Thermal Zoning on Building Energy Simulations: Ground-Truth Validation and Sensitivity Analysis

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submitted by
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Declaration of Authorship

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I further declare that I have not submitted the thesis in the same or similar form to any other examination authority.

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Declaration of generative AI use

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Abstract

Urban building energy models lack information about dwelling layouts and stairwells, although thermal zoning influences energy, load, and comfort-related simulation results. This thesis evaluates the Multizone Assignment Algorithm for generating dwelling–stairwell thermal zoning for multi-family residential buildings from limited geometric and auxiliary input data.

The methodology combines geometric and topological validation with a zoning-variant sensitivity analysis. Two floor plan datasets are converted into a common ground-truth representation and compared with algorithm-generated zoning using validation metrics and diagnostic classifications. Zoning variants represent changes in zoning granularity, internal boundary placement, stairwell representation, and thermal treatment. Their effects on annual heating demand, peak heating load, and overheating hours are assessed in a TEASER–AixLib workflow under varying physical, operational, occupancy-related, and weather conditions.

The results show that the algorithm reproduces the principal dwelling–stairwell structure. Less reliable outputs are associated with footprint geometry, inferred stairwell organisation, and estimated dwelling subdivision. The findings support its use as a scalable zoning approximation where floor plans are unavailable.

Keywords: automated thermal zoning, Multizone Assignment Algorithm, building energy simulation, ground-truth validation, zoning-variant sensitivity analysis, multi-family residential buildings

Zusammenfassung

Urbane Gebäudeenergiemodelle enthalten häufig keine Informationen über Wohnungsgrundrisse und Treppenhäuser, obwohl die thermische Zonierung energie-, last- und komfortbezogene Simulationsergebnisse beeinflusst. Diese Arbeit bewertet den Multizone Assignment Algorithm zur Erzeugung einer wohnungs- und treppenhausbasierten thermischen Zonierung für Mehrfamilienwohngebäude auf Grundlage begrenzter geometrischer und zusätzlicher Eingabedaten.

Die Methodik verbindet eine geometrische und topologische Validierung mit einer Sensitivitätsanalyse der Zonierungsvarianten. Zwei Grundrissdatensätze werden in eine gemeinsame Ground-Truth-Darstellung überführt und anhand von Validierungskennwerten und diagnostischen Klassifikationen mit der algorithmisch erzeugten Zonierung verglichen. Die Zonierungsvarianten bilden Änderungen der Zonierungsgranularität, der Lage interner Grenzen, der Treppenhausdarstellung und der thermischen Behandlung des Treppenhauses ab. Ihre Auswirkungen auf den jährlichen Heizwärmebedarf, die maximale Heizlast und die Überhitzungsstunden werden in einem TEASER–AixLib-Workflow unter variierenden physikalischen, betrieblichen, belegungsbezogenen und klimatischen Bedingungen untersucht.

Die Ergebnisse zeigen, dass der Algorithmus die wesentliche Wohnungs–Treppenhaus-Struktur reproduziert. Weniger zuverlässige Ergebnisse stehen mit der Grundrissgeometrie, der abgeleiteten Treppenhausorganisation und der geschätzten Wohnungsunterteilung in Zusammenhang. Die Ergebnisse unterstützen seinen Einsatz als skalierbare Zonierungsnäherung, wenn keine Grundrisse verfügbar sind.

Schlüsselwörter: automatisierte thermische Zonierung, Multizonen-Zuweisungsalgorithmus, Gebäudesimulation, Ground-Truth-Validierung, Sensitivitätsanalyse von Zonierungsvarianten, Mehrfamilienhäuser

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List of Symbols

Symbol	Description	Unit
A_{dw}	Observed mean dwelling area	m^2
$\hat{A}_{\mathrm{dw}}$	Estimated mean dwelling area	m^2
$\hat{A}_{\mathrm{dw},r}$	Estimated mean dwelling area in regime r	m^2
A_{fp}	External building footprint area	m^2
α_r	Intercept of the piecewise linear model in regime r	m^2
β_r	Slope of the piecewise linear model in regime r	1
I_r	Footprint-area interval associated with regime r	m^2
D	Rasterisation resolution	dpi
n	Drawing-scale denominator	1
s_{scale}	Pixel-to-metre conversion factor based on the drawing scale	$\mathrm{m\,px}^{-1}$
s_{ref}	Pixel-to-metre conversion factor based on a reference length	$\mathrm{m\,px}^{-1}$
s_{area}	Pixel-to-metre conversion factor based on a reference area	$\mathrm{m\,px}^{-1}$
L_{real}	Known real-world reference length	m
d_{px}	Measured reference length in the raster image	px
A_{real}	Known real-world reference area	m^2
A_{px}	Corresponding digitised polygon area in the raster image	px^2
C_{ij}	Assignment cost between ground-truth zone i and predicted zone j	1
G_i	Ground-truth zone i	-
P_j	Predicted zone j	-
N_{GT}	Number of ground-truth zones	1
N_{pred}	Number of predicted zones	1
N_{matched}	Number of valid matched ground-truth–prediction zone pairs	1
N_{FN}	Number of unmatched ground-truth zones	1
N_{FP}	Number of unmatched predicted zones	1

ΔN	Zone-count deviation	1
IoU_{ij}	Intersection over union between ground-truth zone i and predicted zone j	1
$\mathrm{IoU}_{\mathrm{overall}}$	Building-level mean intersection over union	1
$\overline{d}$	Building-level mean centroid distance across valid matched zone pairs	m
$\overline{\mathrm{AreaError}}$	Building-level mean relative area error across valid matched zone pairs	%
E_{GT}	Set of ground-truth dwelling–stairwell adjacency relations	-
E_{pred}	Set of predicted dwelling–stairwell adjacency relations	-
E_{missing}	Set of ground-truth adjacency relations missing from the predicted layout	-
E_{extra}	Set of additional adjacency relations in the predicted layout	-

List of Abbreviations

AI	artificial intelligence
BSP	Binary Space Partitioning
CityGML	City Geography Markup Language
CPZ	Convex Partition Zoner
DACH	Germany, Austria, and Switzerland
DLR	German Aerospace Center
Energy ADE	Energy Application Domain Extension
GIS	geographic information system
HVAC	heating, ventilation, and air conditioning
IoU	intersection over union
LoD	Level of Detail
LPG	LoadProfileGenerator
MSD	Modified Swiss Dwellings
MZA	Multizone Assignment Algorithm
OSM	OpenStreetMap
PDF	Portable Document Format
RC	resistance–capacitance
UBEM	urban building energy modelling
VDI	Verein Deutscher Ingenieure
WWR	window-to-wall ratio
YOC	year of construction

1 Introduction

Buildings account for a substantial share of global energy consumption and greenhouse-gas emissions (Cabeza et al., 2022). Improving the energy performance of the existing building stock is therefore an important part of the transition towards climate-neutral cities. Building energy simulation supports this transition by allowing researchers and practitioners to evaluate renovation strategies, heating demand, peak heating loads, and indoor thermal conditions before measures are implemented (Østergård et al., 2016). However, urban energy planning requires these assessments not only for individual buildings but also for neighbourhoods, districts, and entire building stocks.

Urban building energy modelling extends building energy simulation to these larger spatial scales and can support renovation planning, decarbonisation scenarios, and energy-policy development (Reinhart and Cerezo Davila, 2016). The reliability of these analyses nevertheless depends on how individual buildings are represented. Urban-scale datasets commonly contain building footprints, heights, roof geometries, construction periods, and use classifications, but rarely include detailed internal layouts. Consequently, information about individual dwellings, stairwells, circulation spaces, and their spatial relationships is often unavailable (Bishop et al., 2024).

The absence of internal building information creates a particular challenge for thermal zoning. Thermal zones define the spatial units in which a building energy model represents temperatures, internal gains, control assumptions, envelope exposure, and heat exchange with neighbouring zones. Previous studies have shown that the selected zoning granularity can influence energy, load, and comfort-related results, while more detailed zoning also increases data requirements, computational effort, and model-preparation time (Johari et al., 2022; Shin and Haberl, 2019). At urban scale, manually reconstructing dwelling- or room-level zones for large numbers of buildings is generally impractical (Wang et al., 2022).

Automated thermal-zone generation offers a way to introduce internal spatial differentiation when detailed floor plans are unavailable. Existing methods generate simplified zoning structures from external building geometry, for example through perimeter–core subdivisions or geometrically driven partitions (Dogan et al., 2016; Xiang et al., 2024). These approaches demonstrate that automated zoning can reduce manual modelling effort, but generic spatial subdivisions are not specifically designed to represent the dwelling–stairwell organisation of multi-family residential buildings.

The Multizone Assignment Algorithm (MZA) addresses this limitation by generating dwelling and stairwell zones from limited geometric and auxiliary building information (Waiz et al., 2025). Within the wider workflow, the resulting zones can be enriched with thermal and operational assumptions and transferred to TEASER–AixLib for dynamic building energy simulation. The method therefore offers a potential connection between limited urban building data and spatially differentiated residential simulation models (Remmen et al., 2018).

The usefulness of such an automated zoning method cannot be established solely by confirming that it generates technically valid polygons. The generated zoning must also represent the principal internal structure required for the intended simulation purpose. An output may be geometrically valid while containing an incorrect number of dwellings, a displaced stairwell, altered dwelling–stairwell relationships, or substantial differences in zone size and position. At the same time, not every geometric difference necessarily produces an equally important simulation effect. A local boundary shift may have little influence on an annual building-level indicator, whereas a change in zoning granularity, heated volume, stairwell treatment, or interzonal heat-transfer pathways may produce clearer consequences (Elhadad et al., 2020).

Previous work demonstrated the feasibility of the Multizone Assignment Algorithm using selected building layouts, but broader evidence on the geometric and topological reliability of its generated dwelling–stairwell zoning remained limited. Existing research also shows that thermal zoning affects building energy simulation, but it does not directly establish how representative deviations associated with an automated dwelling–stairwell zoning workflow influence different performance indicators. Evaluating the method therefore requires two connected perspectives: whether the generated zoning corresponds to independent reference structures and whether representative differences in that zoning are relevant to the simulation outputs.

Against this background, this thesis evaluates the suitability of the Multizone Assignment Algorithm for generating simulation-relevant dwelling–stairwell thermal zoning for small- to medium-sized multi-family residential buildings from limited geometric and auxiliary input. The study combines geometric validation with a zoning-variant sensitivity analysis. The geometric validation evaluates the number, geometry, position, and dwelling–stairwell adjacency of the zones generated by the algorithm. The simulation analysis then examines how representative zoning-deviation mechanisms influence annual heating demand, peak heating load, and overheating hours under uncertain simulation conditions.

The validation uses two independent floor plan datasets to construct dwelling–stairwell reference zoning from different forms of source information. Separate extraction workflows convert both datasets into a common ground-truth representation, while the input-preparation workflow derives only the external footprint and auxiliary information required by the Multizone Assignment Algorithm. The generated and reference zones are then aligned, matched, and assessed using geometric and topological metrics, including zone-count deviation, intersection over union, area error, centroid distance, and dwelling–stairwell topology. A hierarchical diagnostic classification combines these measures to distinguish good and moderate outcomes from characteristic deviation-related cases and thereby evaluates both the overall reliability of the algorithm and the mechanisms behind less reliable zoning outputs.

The zoning-variant sensitivity analysis translates the main deviation mechanisms identified through validation into controlled zoning representations, including changes in zoning granularity, internal boundary placement, stairwell representation, and stairwell thermal treatment. The variants are implemented in a TEASER–AixLib (Remmen et al., 2018) simulation workflow and compared using annual heating demand, peak heating load, and overheating hours. To reflect the uncertainty of limited-information building models, the analysis varies physical, operational, occupancy-related, and weather assumptions and examines whether zoning-related differences remain visible across these conditions.

The thesis begins by reviewing the theoretical background and existing research on urban building energy modelling, thermal zoning, automated zone generation, the Multizone Assignment Algorithm, reference floor plan data, and simulation uncertainty. This review identifies the research gaps concerning the geometric and topological reliability of algorithm-generated dwelling–stairwell zoning and the simulation relevance of representative zoning deviations. To address these gaps, the study develops a research method that combines independent geometric validation with a zoning-variant sensitivity analysis. The resulting validation and simulation outcomes are then presented and interpreted in relation to the existing literature, the study objectives, and the requirements of scalable residential building energy modelling. The thesis concludes by synthesising the main findings, contributions, limitations, and opportunities for improving the transparent and reliable use of automated thermal zoning.

2 Point of departure

This chapter establishes the theoretical and research background for evaluating automated thermal zoning in building energy simulation. It first discusses thermal zoning as a modelling decision and explains how zoning granularity affects the balance between simulation detail, data requirements, and modelling effort. The chapter then examines the limitations of geometric data in urban building energy modelling and reviews automated approaches for generating thermal zones from limited input data, with particular focus on the Multizone Assignment Algorithm. Since the thesis also evaluates the simulation relevance of zoning deviations, the chapter discusses simulation uncertainty and sensitivity analysis. It concludes by identifying the research gaps and deriving the research framework.

2.1 Urban building energy modelling

Urban building energy modelling extends building energy modelling from individual buildings to groups of buildings, districts, or entire cities, enabling estimation of energy demand at larger spatial scales. Reinhart & Cerezo Davila (2016) describe urban building energy modelling as a bottom-up modelling approach that applies physical building energy models to groups of buildings to estimate operational energy use at neighbourhood or city scale.

A central limitation of urban building energy modelling is the restricted availability of detailed building information. Urban-scale datasets usually provide only high-level geometric and semantic information, such as building footprints, heights, numbers of storeys, use types, or years of construction, rather than the detailed input that individual building simulation requires. Chen & Hong (2018) note that detailed 3D geometry and thermal zoning are difficult to obtain for each building at urban scale.

Therefore, typical urban building energy modelling geometry inputs often consist only of geographic information system (GIS)-based building geometry.

City Geography Markup Language (CityGML) and related semantic 3D city models provide an important basis for organising urban-scale building geometry. As illustrated in Fig. 2.1, CityGML represents buildings at different levels of detail, ranging from footprint representations to detailed exterior and interior building models. LoD2 is particularly relevant for urban building energy modelling because it is commonly available for large urban datasets and provides the external building volume and roof geometry that envelope-based simulation requires.

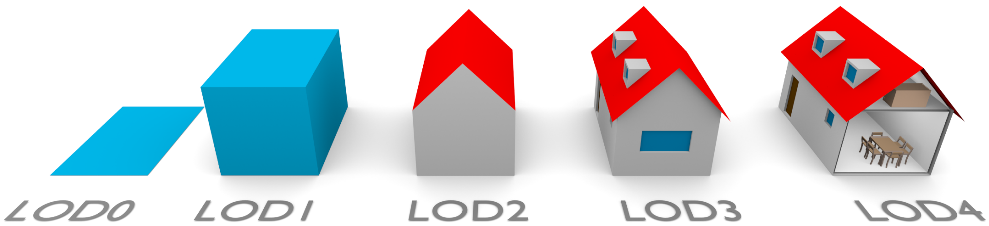


Fig. 2.1: CityGML building representations from a two-dimensional footprint in LoD0 to an interior building model in LoD4. Reproduced from Biljecki et al. (2016).

Higher levels of detail, especially LoD3 and LoD4, can include façade openings and interior structures, but such information is rarely consistently available at the district or city scale. As a result, LoD2-based models still typically require additional methods to infer the internal spatial divisions that define thermal zone boundaries (Biljecki et al., 2016). Although Agugiaro et al. (2018) show that Energy Application Domain Extension (Energy ADE) can represent simulation-relevant building information, the absence of internal layouts still requires automated zoning methods.

LoD2-based urban datasets often lack key thermal simulation attributes, such as construction age, retrofit state, envelope properties, window areas, ventilation, schedules, internal gains, and setpoints. Therefore, simulation-ready models require both spatial enrichment through thermal zoning and typological enrichment of thermal and operational parameters.

TEASER generates reduced-order thermal models from available geometry and typological assumptions, thereby enriching limited urban input data for simulation

(Remmen et al., 2018). However, such workflows still require an earlier zoning step to define the thermal zones. Since urban building energy modelling applies building energy modelling to large numbers of buildings, manually defining missing thermal zone structures is not scalable. Automated zoning methods must therefore infer approximate apartment, corridor, and stairwell structures when urban-scale datasets do not provide them.

2.2 Thermal zones as modelling units

Thermal zoning is a primary modelling decision in building energy simulation because it divides a building's internal space into heat-balance units. Shin & Haberl (2019) describe a thermal zone as a space or group of spaces with sufficiently similar heating and cooling requirements, thermal behaviour, usage patterns, and control requirements for one control device, such as a thermostat or temperature sensor, to regulate them. Therefore, a thermal zone is not only a geometric subdivision but also a heat-transfer and control concept. Merging spaces with different thermal behaviours into a single zone reduces these differences to one common zone condition. The zoning decision therefore affects how the simulation model represents zone-level thermal and operational conditions. Johari et al. (2022) use a similar definition in urban building energy modelling, where a thermal zone refers to a portion of a building with sufficiently similar HVAC demand for a single controlling unit or sensor to regulate it.

2.2.1 Zoning granularity and simulation purpose

Zoning granularity describes the spatial level at which the simulation model divides a building into thermal zones, depending on the simulation purpose, available data, and required accuracy. Reducing the number of zones is a common simplification strategy, but researchers must evaluate this strategy because it affects modelling reliability, computational effort, and simulation accuracy (Johari et al., 2022). Elhadad et al. (2020) similarly show that lower zoning detail can reduce simulation effort, but strong aggregation may cause deviations in heating, cooling, and comfort-related results.

Jansen et al. (2021) also demonstrate this trade-off by comparing different zoning strategies using a TEASER-based reduced-order model and an EnergyPlus reference model. They show that suitable zoning strategies can keep annual heating and cooling demand within approximately $\pm 10\,\%$ while reducing computational effort by up to $97\,\%$. In contrast, unsuitable discretisation may produce unrealistic local thermal behaviour. Kühn et al. (2024) further relate zoning granularity to the distinction between high-order models and reduced-order models: high-order models provide greater spatial detail, while reduced-order models simplify the physical representation and aggregate zones, for example, at a flat or whole-building level.

In automated zoning workflows, zoning granularity can also deviate from the intended internal structure when the algorithm merges, oversubdivides, or displaces dwellings or represents shared circulation areas differently. Such deviations may affect simulation outputs by altering façade allocation, interzonal heat-transfer areas, zone temperatures, or the treatment of unheated shared spaces (Brès et al., 2017).

2.2.2 Zoning levels

Building energy simulation can use different levels of thermal zoning, ranging from detailed room-level models to highly aggregated single-zone models. The appropriate zoning level depends on the simulation purpose, available input data, and acceptable modelling effort. Table 2.1 summarises the main zoning levels that the literature discusses, together with their typical purposes, limitations, and supporting references.

These findings show that each zoning level serves a distinct simulation purpose and entails different limitations. In the context of urban building energy modelling, this poses a specific challenge: more detailed dwelling- and stairwell-level zoning may improve the representation of multi-family residential buildings, but such zoning requires internal spatial information that urban-scale datasets usually do not provide. This gap motivates the need for automated thermal zone generation methods that can infer suitable zoning structures from limited geometric input.

Tab. 2.1: Thermal zoning levels identified in the literature.

Zoning level	**Description**	**Simulation purpose**	**Limitation**	**References**
Room level	The model represents each room as an individual zone.	Individual building simulation when detailed input is available.	Requires internal layouts and high modelling effort.	Kühn et al. (2024); Elhadad et al. (2020)
Dwelling/flat level	The model represents each flat as a single zone.	Individual- or district-scale simulation of multi-family buildings.	Does not represent room-level variation within each dwelling.	Kühn et al. (2024)
Storey level	The model represents each floor as a single zone.	Simplified building, city, or district simulations with vertical differences.	Cannot distinguish between different dwellings on the same floor.	Johari et al. (2022); Elhadad et al. (2020)
Single-zone	The model represents the entire building as a single zone.	Large-scale neighbourhood, city, or building-stock simulations.	Averages differences between dwellings, orientations, floors, and shared spaces.	Johari et al. (2022); Elhadad et al. (2020)
Perimeter–core	The model divides the building into perimeter and core zones.	Early-stage design or automated urban modelling.	Does not represent the internal structure of multi-family buildings.	Shin & Haberl (2019); Dogan et al. (2016)

2.3 Automated thermal zone generation

Researchers have developed several automated approaches to generate internal spatial or thermal zone representations when detailed building information is unavailable. Some methods focus on the procedural generation of interior layouts, such as Lopes et al. (2010), who generate room layouts for different building types and multi-storey buildings. In building energy modelling, however, automated methods commonly use simplified internal subdivisions to define suitable simulation zones from limited geometric information. For this purpose, Dogan et al. (2016) propose Autozoner, which subdivides building massing models into perimeter and core thermal zones when available data do not specify the interior spaces. More recently, Xiang et al. (2024) propose the Convex Partition Zoner (CPZ), which automatically generates perimeter and core zones from building floor geometries when available data do not specify the actual HVAC zones or designers have not yet defined them.

At the urban scale, automated model-generation workflows also use limited geospatial data to produce simulation-ready energy models. For example, Cerezo Davila et al. (2016) generate a citywide urban building energy model for Boston from existing geospatial datasets and archetype libraries, while Chen et al. (2017) introduce CityBES for automatically generating and simulating urban building energy models from city datasets.

For multi-family residential buildings, generic perimeter–core or floor-level zoning does not fully capture the dwelling and stairwell organisation that shapes dwelling-level thermal behaviour, interzonal heat transfer, and user profiles. Recent work by Waiz et al. (2025) addresses this limitation through the Multizone Assignment Algorithm, which generates dwelling- and stairwell-based thermal zone structures from limited geometric and auxiliary inputs within the TEASER–AixLib simulation framework. The following section therefore discusses the Multizone Assignment Algorithm in more detail.

2.4 Multizone Assignment Algorithm

The objective of the Multizone Assignment Algorithm is to infer an approximate dwelling- and stairwell-based thermal zoning from limited geometric and semantic input. This is particularly relevant for multi-family residential buildings and dwelling blocks, where a single-zone representation can mask differences between individual dwellings and shared access spaces. These differences are relevant because unheated neighbouring zones and temperature differences between dwellings can cause interzonal heat exchange and influence heating-demand and peak-load predictions (Kühn et al., 2024). Occupancy-related internal gains and user-controlled setpoint temperatures introduce further variation in residential building-energy and heating-load predictions (Ioannou and Itard, 2015; Pflugradt et al., 2022).

At the level relevant for this thesis, the important output is not detailed internal floor-plan geometry but a thermal zoning that divides the floor area into dwelling and stairwell zones. This distinction is important for the validation approach of the present thesis because the validation assesses the generated zones at the dwelling–stairwell level rather than at the level of individual rooms or architectural details.

For the present thesis, a distinction is required between the complete simulation workflow and its spatial zone generation method. The full workflow includes data aggregation, preprocessing, automatic zoning, thermal enrichment, and dynamic simulation in TEASER–AixLib. The geometric validation in this thesis focuses on the spatial output of the zoning step. It evaluates the number, geometry, location, and dwelling–stairwell adjacency of the zones that the algorithm generates.

2.4.1 Workflow structure

The workflow comprises three conceptual layers: data aggregation and preprocessing, automatic zoning, and enrichment and simulation. The data aggregation and preprocessing layer prepares the geometric, semantic, and typological information that the zoning process requires. The enrichment and simulation layer then translates the zones that the automatic zoning layer generates into thermal simulation models. Both layers depend strongly on available regional data, typological assumptions, and modelling choices. By contrast, the automatic zoning layer defines the spatial logic for stairwell placement and can be examined separately once the preceding layer supplies the zoning inputs. Figure 2.2 provides a conceptual overview of the Multizone Assignment Algorithm workflow.

Data aggregation and preprocessing

The data aggregation and preprocessing stage prepares the building information required by the automatic zoning process. In the original Multizone Assignment Algorithm workflow, the preprocessing layer combines geodata from OpenStreetMap (OSM) (OpenStreetMap contributors, 2024) with CityGML data to create a more

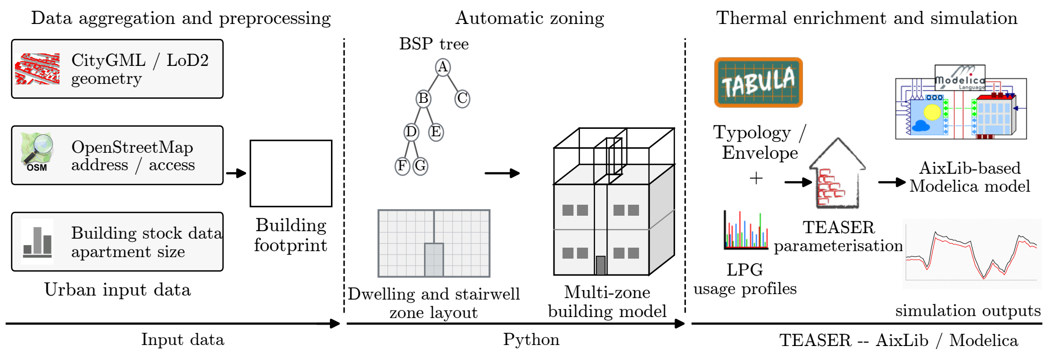


Fig. 2.2: Conceptual overview of the Multizone Assignment Algorithm workflow from limited urban input data through automatic zoning to thermal simulation.

complete description of the investigated buildings. CityGML provides the LoD2-type external building geometry, while OpenStreetMap supplements this information with attributes such as building type, roof type, and building address points.

In addition to CityGML and OpenStreetMap, the Multizone Assignment Algorithm workflow uses external building-stock information to populate attributes that urban geometry data typically lack. This additional dataset draws on building-property classifications derived from German census data, as reported by Blanco et al. (2023). It enriches the building data with construction-year typologies and related building-stock attributes. The wider Multizone Assignment Algorithm workflow then uses these attributes to support assumptions about construction type, envelope properties, average dwelling size, and the expected number of dwelling units.

The rule-based determination of the stairwell location forms another relevant part of the preprocessing stage. The Multizone Assignment Algorithm workflow estimates the stairwell location from available external information, such as the building address point or street layout. It then orients the stairwell orthogonally to the nearest suitable external wall. This step is important because the stairwell can represent an unheated shared zone that influences heating demand, while its position defines an access boundary that later constrains the automatic generation of dwelling zones. The workflow estimates the stairwell dimensions from the floor area and applicable stair standards. However, the authors note that future refinements should account more explicitly for construction year and building type.

This part of the workflow depends strongly on the availability and quality of external geodata, address information, regional building-stock assumptions, and assumptions about stairwell placement.

Automatic zoning

The automatic zoning stage uses the prepared inputs to generate dwelling and stairwell zones. The framework offers several zoning options, including manual zoning, perimeter–core zoning, graph-based clustering, and a heuristic automatic zoning algorithm. However, the original study focuses primarily on the heuristic approach for generating dwelling zones in multi-family buildings.

In the heuristic approach, Binary Space Partitioning (BSP) first discretises the external building footprint by recursively subdividing the base surface into smaller spatial elements up to a predefined granularity (Fuchs et al., 1980). As illustrated in Fig. 2.3, each subdivision creates new child nodes in a tree structure until the process reaches the final partitions. These final partitions, referred to as leaves, form the basic spatial units for the subsequent zone assignment.

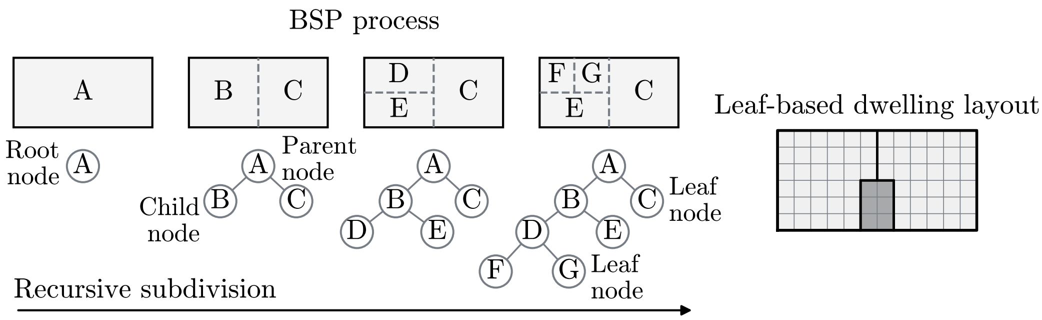


Fig. 2.3: Binary Space Partitioning in the Multizone Assignment Algorithm workflow. The process recursively subdivides the floor area into a tree structure whose leaf nodes form the spatial units for zone assignment.

After Binary Space Partitioning discretises the floor area, the workflow assigns additional spatial information to the partition leaves. This information includes leaf adjacency, contact with the external building boundary, contact with the stairwell or access boundary, and barrier edges that the assignment procedure must not cross. The automatic zoning procedure requires these relations because it does not generate independent free-form polygons. Instead, it generates candidate dwelling regions within the residential floor area and subsequently assigns the discretised leaves to these regions. The algorithm does not treat the stairwell as a dwelling seed. Instead, the stairwell position established during preprocessing defines the access boundary from which the algorithm generates the residential sub-areas.

In the version of the Multizone Assignment Algorithm evaluated in this thesis, Voronoi seed placement forms the basis of the heuristic zoning logic (Okabe et al., 2000). For the specified number of dwelling zones, the algorithm places seed points along the boundary between the stairwell and the remaining residential floor area. Each seed represents one dwelling zone per floor. The preceding data aggregation and preprocessing layer determines the number of dwelling zones from auxiliary information, such as the average dwelling size or the expected number of households per building volume (Waiz et al., 2025).

The algorithm tests several seed configurations iteratively. Depending on the access geometry, it places the seeds either along one access boundary or across the two most relevant access boundaries. The initial configurations use regular spacing, while later iterations introduce random variation to explore alternative dwelling-zone layouts.

For each seed configuration, the algorithm generates a Voronoi diagram and clips it to the remaining residential region while excluding the stairwell and barrier edges. Each clipped cell represents a candidate dwelling region. The algorithm accepts a candidate solution only when all cells satisfy the principal spatial constraints, including sufficient contact with the access boundary and external façade, and when the solution contains no empty or invalid cells. Among the valid candidates, the algorithm selects the configuration that provides the best balance between compact zone geometry and similar dwelling areas.

After selecting the best polygon-level candidate, the algorithm assigns the partition leaves to the Voronoi cells according to their largest intersection area. If this criterion does not produce an assignment, the procedure uses the nearest seed or a representative point as a fallback. The subsequent post-processing stage improves zone connectivity, removes local geometric irregularities, and preserves the principal spatial constraints. The algorithm accepts these modifications only when the resulting zoning continues to satisfy the main constraints. Overall, the method combines Binary Space Partitioning discretisation, Voronoi-based candidate generation, objective-based selection, and local post-processing to generate connected dwelling-level thermal zones around the stairwell geometry from the preprocessing stage. Because the final boundaries follow combinations of discrete partition leaves and may be locally adjusted during post-processing, the generated zones can differ from irregular architectural boundaries even when they preserve the principal dwelling–stairwell organisation.

Enrichment and simulation

After the automatic zoning stage, the Multizone Assignment Algorithm workflow converts the generated dwelling and stairwell zones into a dynamic building energy simulation model. This stage combines TABULA, TEASER, AixLib, Modelica, and VDI 6007-1, which fulfil different functions within the simulation workflow.

TABULA is a European residential building-typology framework that classifies buildings according to characteristics such as country, building type, construction period, and renovation state (Loga et al., 2016). The typologies provide representative construction and envelope properties when detailed building-specific information is unavailable. In the Multizone Assignment Algorithm workflow, the German TABULA typologies support the assignment of construction-period classes, material and envelope properties, and window-related assumptions.

TEASER is an open-source Python framework for the automated generation and parameterisation of reduced-order building energy models (Remmen et al., 2018). It combines the generated thermal-zone geometry with typological, thermal, and operational information to prepare a simulation-ready building model. In the Multizone Assignment Algorithm workflow, TEASER receives the generated dwelling and stairwell zones and parameterises the corresponding thermal model in accordance with VDI 6007-1.

VDI 6007-1 defines a reduced-order approach for representing the transient thermal behaviour of rooms and buildings (VDI, 2015). The resulting models approximate heat storage and heat transfer with a reduced number of thermal states, thereby supporting dynamic simulation with lower computational effort than more detailed physical models.

The resulting thermal model uses components from AixLib, an open-source Modelica library for buildings, building services, and energy systems (Müller et al., 2016). Modelica is an equation-based modelling language used to describe and simulate dynamic physical systems (Modelica Association, 2026). TEASER therefore generates and parameterises the building model, while AixLib provides the Modelica components used to represent its thermal behaviour. The resulting Modelica model can then be executed using a compatible simulation environment.

These models represent thermally conductive building components using resistance–capacitance (RC) elements, which describe heat storage and heat transfer through building elements with reduced computational effort. Since the Multizone Assignment Algorithm explicitly generates several neighbouring zones, interzonal heat transfer becomes relevant. The workflow therefore extends the reduced-order model with additional thermal connections between adjacent zones, thereby representing heat exchange among dwellings, floors, and unheated stairwell zones (Groesdonk et al., 2023; Waiz et al., 2025).

This simulation layer also assigns internal gains and usage profiles to the zones. In addition to standardised usage assumptions, the Multizone Assignment Algorithm workflow can use LoadProfileGenerator (LPG) profiles (Pflugradt et al., 2022) to describe occupants, appliances, and lighting through more detailed household-specific behaviour. The workflow can then distribute these profiles across the dwelling zones so that different dwellings receive different occupancy and internal-gain schedules.

2.4.2 Evaluation of the Multizone Assignment Algorithm workflow

Waiz et al. (2025) show that the Multizone Assignment Algorithm generates consistent zoning outputs for several building floor plans in their study and supports more detailed heating load prediction than a single-zone model. However, this additional spatial detail increases computational effort, and the multi-zone model that includes interzonal heat transfer requires up to 750 % more simulation time than the single-zone model.

The authors identify further refinement of the zoning method, extensive validation using real buildings, and improved treatment of building-stock and construction-period assumptions as important directions for future work (Waiz et al., 2025).

These findings motivate a multilevel evaluation of the Multizone Assignment Algorithm. Geometric validation determines whether the dwelling and stairwell zones that the algorithm generates correspond to realistic internal building structures by assessing both geometric and topological agreement. However, geometric comparison alone cannot determine whether the observed zoning deviations affect simulation outputs.

A zoning layout that the algorithm generates may differ from the reference floor plan structure. However, the importance of this difference depends on whether it changes performance indicators such as heating demand, peak heating load, or overheating. Therefore, evaluating the Multizone Assignment Algorithm requires both an assessment of its geometric and topological reliability and an investigation of the simulation relevance of representative zoning deviations. The following section reviews simulation uncertainty and sensitivity analysis as the basis for comparing alternative zoning representations under varying input conditions.

2.5 Simulation uncertainty and sensitivity analysis

Building energy simulation models rely on assumptions about geometry, construction, operation, occupancy, and system behaviour. Since these inputs often remain incomplete or uncertain, simulation outputs should not be interpreted solely on the basis of a single fixed model configuration. Sensitivity analysis provides a systematic way to examine how changes in modelling assumptions or model representations affect selected simulation outputs. In the context of automated thermal zoning, the zoning representation constitutes the modelling factor of interest, while physical, operational, occupancy-related, and weather inputs define the varying conditions under which the zoning variants are compared. This section discusses zoning variants as a modelling choice, introduces input uncertainty in urban building energy simulation, and explains how alternative modelling variants can be evaluated under uncertain input conditions.

2.5.1 Zoning variants as a modelling choice

As discussed in Sec. 2.2, thermal zoning affects how building energy simulation represents heat transfer, control assumptions, and spatial temperature differences. Studies of multi-family residential buildings show that zoning detail and interzonal heat exchange affect heat-demand predictions and model accuracy. Different dwelling-level temperature setpoints can cause heat exchange between flats and floors, while an unheated stairwell can influence the heat demand of the building and its individual zones (Kühn et al., 2024; Waiz et al., 2025).

However, the Multizone Assignment Algorithm-based workflow introduces a specific source of modelling uncertainty. In this case, the algorithm automatically generates the zoning structure from limited geometric and auxiliary input rather than relying solely on a manually selected level of detail. The resulting zoning structure may therefore deviate from reference dwelling and stairwell structures. Such deviations can include under-zoning, over-zoning, alternative dwelling–stairwell configurations, displaced stairwell locations, and different representations of stairwell spaces.

Geometric validation alone cannot determine whether such deviations meaningfully affect simulation outputs. Evaluating automated zoning methods therefore also requires examining whether zoning-related differences remain visible when the remaining simulation inputs vary within plausible ranges.

2.5.2 Input uncertainty in urban building energy simulation

Building energy simulation depends strongly on the quality and completeness of input data. This issue is especially relevant in urban building energy modelling, where datasets often lack complete building-specific information and typological or statistical enrichment methods must estimate the missing values. Malhotra et al. (2022) show that attributes such as year of construction (YOC), building usage, and refurbishment state represent important enrichment inputs for CityGML-based heating demand modelling.

Such enrichment introduces input uncertainty because the values assigned by the enrichment procedure may not correspond exactly to the real building. Prataviera et al. (2022) show that input uncertainty can affect urban building energy simulation results and argue that uncertainty and sensitivity analyses are needed to identify the assumptions that most strongly influence energy-demand predictions. In the Multizone Assignment Algorithm-based workflow, this issue is relevant because the generated zoning structure is combined with uncertain enrichment inputs such as construction age, retrofit state, envelope properties, ventilation assumptions, setpoints, internal gains, and occupancy profiles.

Weather and climate data form an additional external input to building energy simulation. Outdoor air temperature and solar radiation affect the building heat balance and therefore influence annual heating demand, peak heating loads, and indoor overheating. Studies comparing alternative weather datasets show substantial differences in simulated heating demand and peak load, while the weather-file construction and represented climate conditions also affect predicted overheating. The selection of the weather location, reference period, and weather-data representation should therefore be treated as a relevant source of simulation uncertainty (Daneshfar et al., 2023; Xie et al., 2025).

These uncertainties are also relevant when alternative zoning representations are compared. A difference observed under one fixed set of construction, operation, occupancy, and weather assumptions may become smaller, larger, or less consistent when these conditions vary. Consequently, conclusions about the relevance of zoning should be based on comparisons across a range of plausible input and boundary conditions rather than on a single deterministic model configuration.

2.5.3 Sensitivity analysis in building performance modelling

In building performance modelling, sensitivity analysis examines how variations in model inputs, assumptions, or representations influence selected simulation outputs. The investigated factors may include numerical inputs, such as ventilation rates or temperature setpoints, as well as discrete modelling choices, such as alternative geometric representations or thermal zoning configurations. The design of the analysis therefore depends on the research objective and on the type of factor being investigated.

Tian (2013) describes sensitivity analysis in building performance analysis as a sequential process that defines the investigated variations, executes the corresponding simulations, and analyses and interprets the resulting outputs. Figure 2.4 summarises this process.

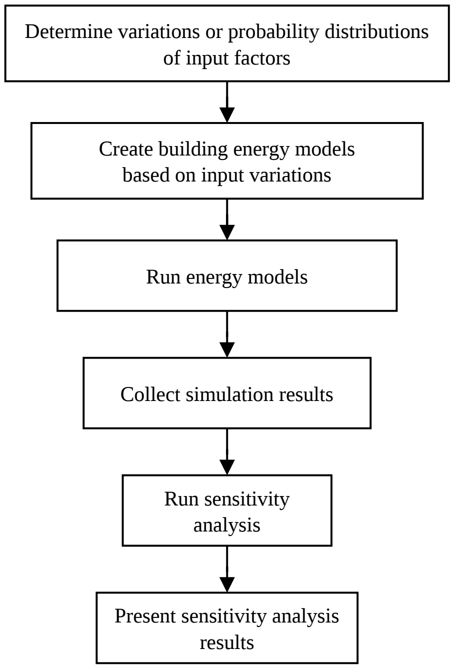


Fig. 2.4: General sensitivity-analysis workflow for building performance analysis. Adapted from Tian (2013).

Sensitivity analysis can therefore be used to investigate both individual input parameters and alternative model configurations. When model configurations are compared, consistent boundary conditions and input assumptions help relate differences in the outputs to the modelling choice being investigated. If the remaining model inputs are uncertain, the configurations can be evaluated across a range of plausible conditions instead of using only one fixed configuration.

2.6 Research gap and framework

Despite these developments, several research gaps remain. First, no previous study has yet systematically validated the zoning geometry generated by the Multizone Assignment Algorithm against independent reference dwelling–stairwell structures.

Since the algorithm infers internal dwelling and stairwell zones from limited external geometry and auxiliary input data, this thesis compares its spatial output with reference ground-truth zoning at the same abstraction level.

Second, existing research does not yet establish whether Multizone Assignment Algorithm-related zoning deviations influence simulation outputs. Such deviations can arise when limited input data lead to an incorrect number of zones, alternative dwelling–stairwell arrangements, shifted stairwell locations, or different stairwell representations. Existing studies show that zoning granularity and interzonal heat transfer can influence building performance simulation. However, they do not directly connect these automated zoning deviations to performance indicators such as heating demand, peak heating load, or overheating.

This thesis addresses these gaps by combining geometric validation with zoning-variant sensitivity analysis. The validation evaluates whether the Multizone Assignment Algorithm can reproduce reference dwelling–stairwell structures, while the simulation analysis examines whether Multizone Assignment Algorithm-related zoning deviations and uncertain enrichment inputs influence selected simulation outputs. The thesis does not validate the complete thermal simulation workflow against measured energy performance but focuses on the zoning structures that the Multizone Assignment Algorithm generates and their relevance for selected simulation outputs.

The central research question of this thesis is:

> To what extent can the Multizone Assignment Algorithm generate simulation-relevant thermal zoning structures for multi-family residential buildings from limited geometric and auxiliary inputs, and how do representative zoning deviations influence selected building energy simulation outputs?

The following two connected research questions address the central research question:

1) **RQ1:** To what extent can the Multizone Assignment Algorithm reproduce dwelling- and stairwell-based reference zoning structures in multi-family residential buildings using limited geometric and auxiliary inputs?

2) **RQ2:** How do Multizone Assignment Algorithm-related zoning deviations identified through geometric validation, such as reduced or increased zoning granularity, alternative dwelling–stairwell configurations, and different stairwell representations, influence selected building performance indicators under uncertain input conditions?

Based on the literature review, the thesis hypothesises that the algorithm can reproduce the dominant dwelling- and stairwell-level zoning structure of multi-family residential buildings with sufficient geometric and topological agreement. It further hypothesises that Multizone Assignment Algorithm-related zoning deviations have a greater influence on peak and comfort-related outputs than on annual heating demand.

3 Research method

This chapter describes the research design used to evaluate the Multizone Assignment Algorithm from geometric and simulation-orientated perspectives. It first presents the validation workflow used to construct independent dwelling–stairwell reference zoning and compare it with the algorithm-generated zones. It then describes the zoning-variant sensitivity analysis used to assess how representative zoning deviations influence the selected simulation outputs under uncertain input conditions.

3.1 Validation

In the standard Multizone Assignment Algorithm workflow, the available input includes the external building geometry but not the internal dwelling and stairwell structure. This validation therefore examines whether the automatically generated zoning reproduces the reference dwelling–stairwell structure with respect to the number, geometry, location, and adjacency of the zones. Assessing these geometric and topological properties requires independent reference zoning.

The broader target context of this validation is multi-family residential building modelling in the Germany–Austria–Switzerland (DACH) region, where automated thermal zoning methods are relevant because urban-scale datasets often lack detailed internal layouts. This regional focus is appropriate because the Multizone Assignment Algorithm targets Central European urban building data and uses enrichment assumptions based on German building-stock information and TABULA typologies (Loga et al., 2016). The validation sample primarily comprises small- to medium-sized multi-family residential buildings because this building range most closely matches the intended application and dwelling–stairwell zoning assumptions of the Multizone Assignment Algorithm.

The validation focuses on the geometric and topological reconstruction capability of the automatic zoning process. It does not evaluate the auxiliary input assumptions, thermal enrichment procedure, or subsequent dynamic simulation stage.

3.1.1 Validation workflow

This thesis conducted the validation at the dwelling–stairwell level. The validation framework treated each dwelling as one thermal zone and each stairwell as a separate zone. This abstraction corresponds to the intended output level of the Multizone Assignment Algorithm and captures the main internal organisation of multi-family residential buildings without requiring room-level detail (Kühn et al., 2024).

The validation required independent reference zoning at the same abstraction level as the algorithm output. Floor plans served as source material for constructing this reference zoning. The study did not treat the original floor plans themselves as ground truth. Instead, the source-specific extraction workflows interpreted and transformed the available spatial information into dwelling and stairwell reference polygons containing their geometries, locations, and spatial relationships.

Figure 3.1 shows the overall validation workflow. The implemented workflow contained two reference-data branches: structured Modified Swiss Dwellings (MSD) data and German real estate floor plan documents (Engelenburg et al., 2025; ImmoScout24, 2026). Each branch used a source-specific procedure to construct dwelling and stairwell reference polygons.

After ground-truth extraction, the two branches were converted into a common reference representation and processed consistently through input preparation, automatic zoning, geometric alignment, zone matching, metric calculation, and diagnostic classification.

The following subsection describes the screening and selection of the reference data sources and explains the methodological challenge of constructing dwelling–stairwell reference zoning from the available floor plan information.

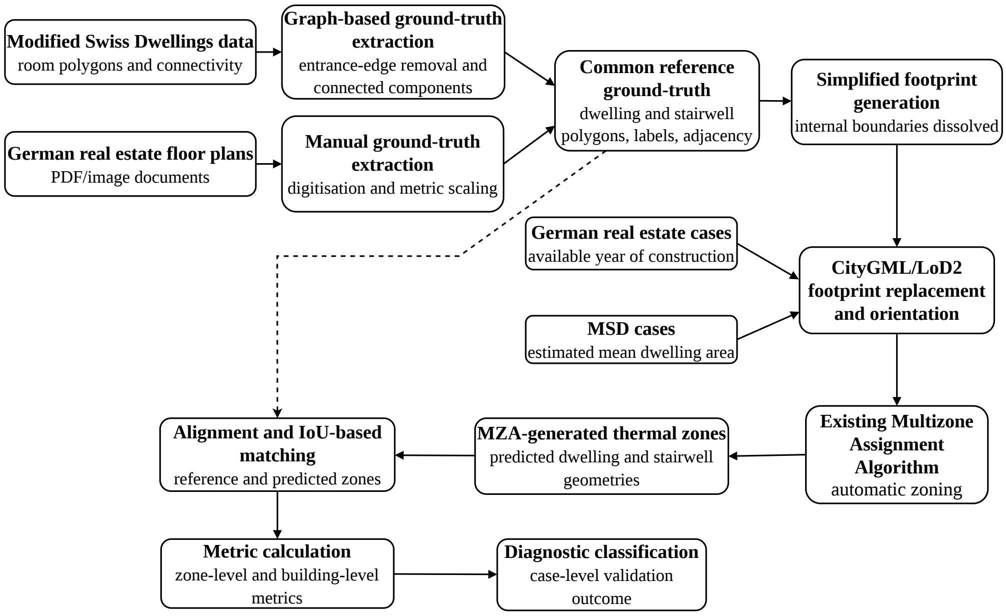


Fig. 3.1: Validation workflow for comparing Multizone Assignment Algorithm-generated zoning with dwelling–stairwell reference zoning.

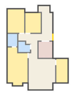
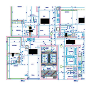
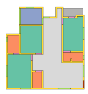
3.1.2 Reference data sources

The validation required floor plan data from which individual dwellings, shared stairwell spaces, their geometries, and their spatial relationships could be identified. The sources also had to contain multi-unit residential layouts and sufficient geometric information to construct a common dwelling–stairwell reference representation.

Several publicly available floor plan datasets were reviewed against these requirements. Table 3.1 summarises their representations and corresponding sources.

Validating the zoning output of the Multizone Assignment Algorithm requires a reference representation of the internal dwelling and stairwell structure. Floor plans provide an important intermediate source of spatial information because they describe the internal organisation of buildings. However, they do not themselves constitute thermal-zone models. Instead, their geometric, semantic, and topological information can be interpreted and transformed into a dwelling–stairwell reference zoning suitable for validation.

Tab. 3.1: Overview of floor plan data sources and their representations. Example images are adapted from the cited sources.

Data source	Example	Representation	Source
RPLAN		Raster-based annotated residential layouts	Wu et al. (2019)
LIFULL		Real estate floor plan images and vectorised subsets	LIFULL & NII (2017)
CubiCasa5K		Raster images with Scalable Vector Graphics polygon annotations	Kalervo et al. (2019)
MLSTRUCT–FP		High-resolution raster floor plans with wall and slab annotations	Pizarro et al. (2023)
Modified Swiss Dwellings		Image, vector, semantic, structural, and graph-based representations	Engelenburg et al. (2025)
ResPlan		Vector-graph representations of residential floor plans	Abouagour & Garyfallidis (2025)
German real estate floor plans		PDF and raster floor plan documents	ImmoScout24 (2026)

The reviewed sources were developed for different purposes. Nevertheless, none of the reviewed sources directly provides a thermal-zoning reference at the dwelling–stairwell abstraction level required for comparison with the Multizone Assignment Algorithm.

This creates a methodological challenge for the validation. The algorithm-generated output cannot be compared directly with the original floor plans. Instead, the source information must first be transformed into a reference representation at the same abstraction level as the algorithm output. This representation must describe the number, geometry, location, and spatial relationships of the dwelling and stairwell zones.

Based on the availability of suitable multi-unit residential layouts and the information required for this transformation, the study selected the Modified Swiss Dwellings dataset and German real estate floor plan documents as complementary reference sources. Table 3.2 summarises the available information and the role of each source in the validation.

Tab. 3.2: Reference data sources used to construct dwelling–stairwell ground truth.

Data source	**Available information**	**Purpose**
Modified Swiss Dwellings	Room polygons, room labels, zone labels, and graph-based connectivity information	Provides a larger structured dataset for reproducible extraction of dwelling and stairwell reference zones.
German real estate floor plans	Floor plan drawings from real building documents, with scale or dimensional information where available	Provides additional validation cases from the German residential building context.

Together, the two sources form a purposive Swiss–German validation sample. The sample supports the evaluation of the Multizone Assignment Algorithm within the defined scope of small- to medium-sized multi-family residential buildings, but it is not interpreted as statistically representative of all European or DACH residential building typologies.

3.1.3 Modified Swiss Dwellings dataset

As introduced in Sec. 3.1.2, the Modified Swiss Dwellings dataset builds on the Swiss Dwellings dataset and provides structured residential floor plan data (Standfest et al., 2022). Its main advantage for this thesis is that it provides not only floor plan images but also polygonal room geometries, semantic room-type labels, zone-type labels, and graph-based connectivity information. These representations enable reproducible derivation of dwelling and stairwell reference zones.

Figure 3.2 summarises the data representations that this study used. The graph-based representation played a particularly important role because it allowed the extraction workflow to infer dwelling boundaries and shared circulation areas from connectivity information rather than from image interpretation alone.

The Modified Swiss Dwellings cases used in this study did not include building-specific metadata such as address information or year of construction. These attributes were not required for the extraction of the reference dwelling–stairwell

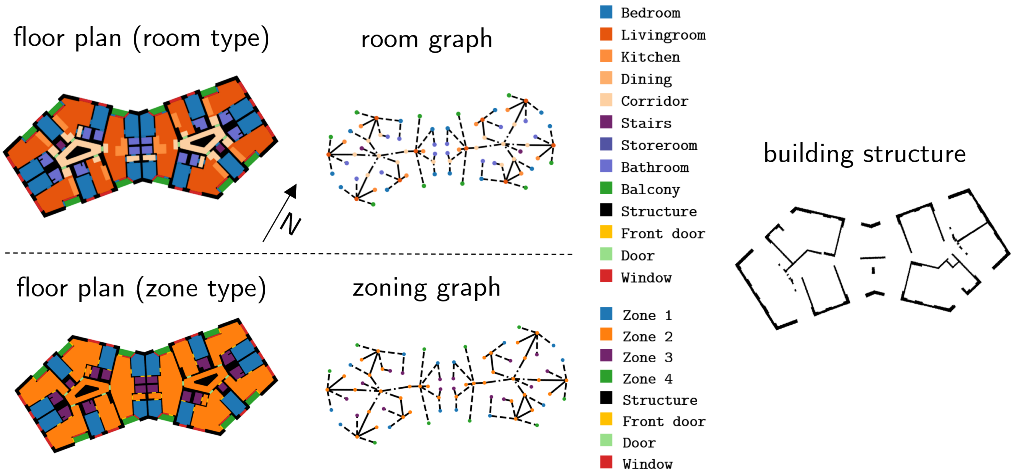


Fig. 3.2: Data representations available in the Modified Swiss Dwellings dataset. Reproduced from Engelenburg et al. (2025).

zoning, which was derived independently from the available room geometries and connectivity information. However, additional auxiliary information was required to prepare the inputs for the Multizone Assignment Algorithm. The missing auxiliary input was therefore estimated using the procedure described in Sec. 3.1.6. This estimated information was used exclusively for the Multizone Assignment Algorithm input-preparation workflow and was not used to define or modify the reference ground-truth zones.

Ground-truth extraction

The Modified Swiss Dwellings dataset provides each floor plan as a graph-based representation. Nodes represent spatial areas and contain polygonal room geometries, semantic room-type labels, and centroid locations. Edges represent spatial connectivity between rooms. The entrance-edge labels played an especially important role in this thesis because they identify transitions between shared circulation spaces and private dwelling interiors.

Figure 3.3 shows the ground-truth extraction workflow. First, the workflow loaded the original room-level graph containing room polygons, room labels, and connections between room. It then removed auxiliary spaces, such as balconies, and deleted the dwelling-entrance edges to separate private dwelling spaces from shared circulation spaces. Next, the workflow split the graph into connected components and classified them as dwellings or stairwell areas. Finally, it merged the room polygons within each component to create the dwelling and stairwell reference zones.

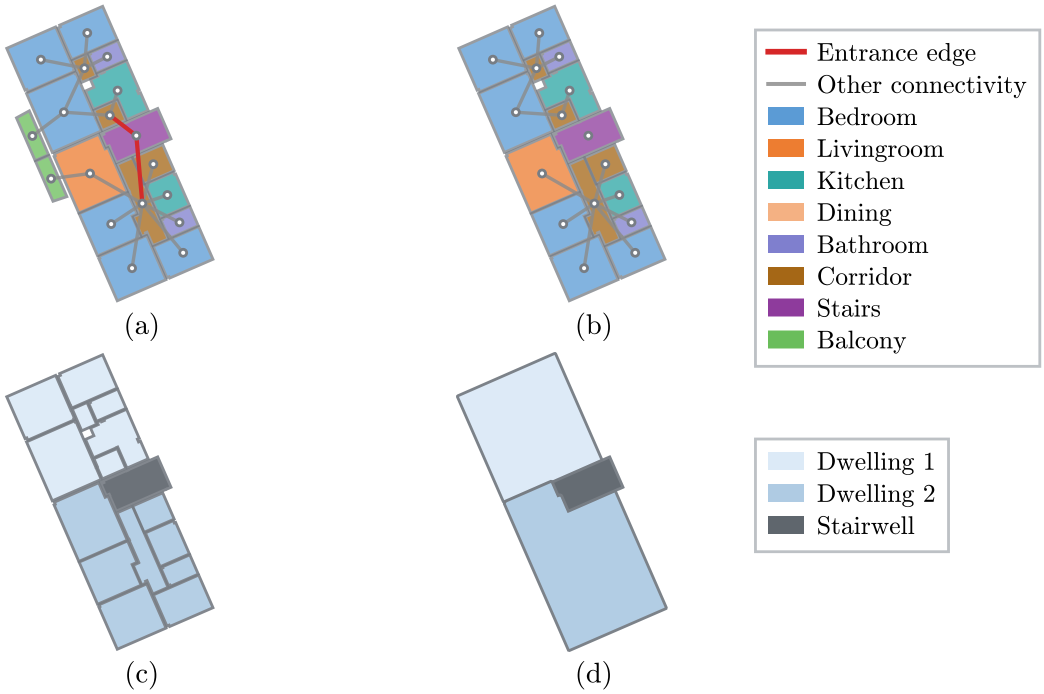


Fig. 3.3: Graph-based extraction of reference ground-truth zones from the Modified Swiss Dwellings dataset. (a) Room-level graph. (b) Graph after removal of auxiliary spaces and entrance edges. (c) Classification of dwelling and stairwell areas. (d) Merged dwelling and stairwell reference geometries.

Buffering operations closed small gaps between adjacent polygons and produced spatially consistent zone outlines. This regularisation reflects the selected dwelling–stairwell abstraction level by deriving thermally relevant reference zones without preserving detailed internal wall thicknesses. Automatic thermal zoning approaches use similar geometric simplifications when detailed interior definitions are unavailable (Dogan et al., 2016).

The filtering procedure retained only cases within the validation scope from the full Modified Swiss Dwellings dataset. Table 3.3 summarises the filtering and final selection process. The manual quality-control process selected 200 floor plans from the eligible cases. Figure A.1 provides a complete overview of the selected layouts.

Tab. 3.3: Selection of Modified Swiss Dwellings cases for validation.

Step	Criterion	Cases
Curated dataset	Cleaned Modified Swiss Dwellings dataset	5372
Scope filter	Buildings with two to six dwellings and one to two stairwells	3678
Manual quality control	Removal of duplicate, ambiguous, unsuitable, or incompatible cases	600
Final validation sample	Selected cases for validation	200

3.1.4 German real estate floor plans

This study used German real estate floor plan documents from publicly available listings on ImmoScout24 (ImmoScout24, 2026) as the second reference source. Unlike the Modified Swiss Dwellings dataset, the real estate listings do not provide these plans as structured graph data. Instead, they provide PDF or image-based drawings. The ground-truth extraction workflow therefore interpreted and digitised the plans before validation.

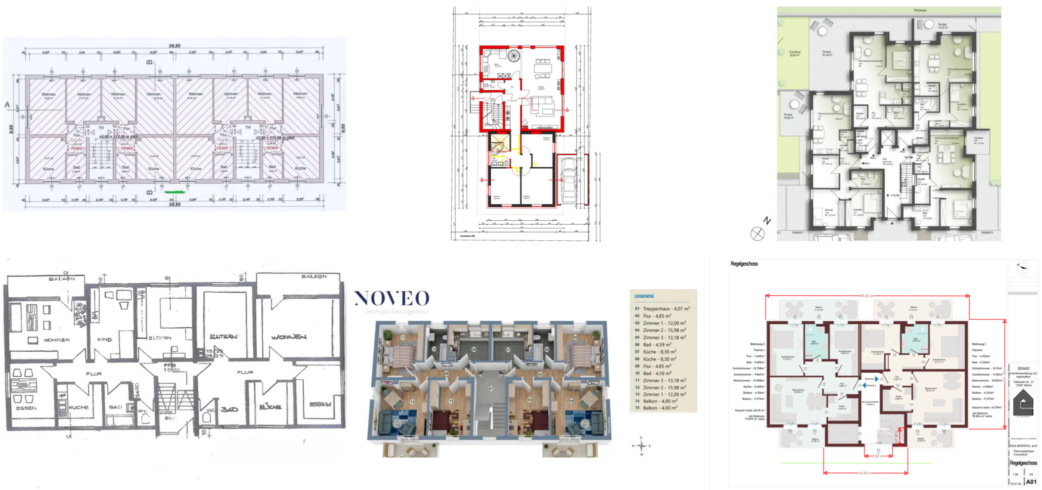


Fig. 3.4: Examples of German real estate floor plan documents used as reference sources.

Figure 3.4 illustrates the range of source-document formats used in the German validation branch. The selection process retained only clearly interpretable multi-family residential layouts with identifiable dwelling and stairwell spaces.

Where available, the scaling procedure used dimensions, scale bars, room areas, or total floor area to convert the digitised drawings into metric geometry. The Multizone Assignment Algorithm input-preparation workflow used the year of construction only as an auxiliary input and not to define the ground-truth polygons. Although the German real estate floor plans do not contain explicit connectivity graphs or semantic room labels, they provide sufficient architectural information for constructing dwelling–stairwell reference zones. They therefore serve as complementary German validation cases.

Ground-truth extraction

Unlike the Modified Swiss Dwellings branch, the German real estate branch started from drawing-based PDF or image floor plans rather than structured graph data. Figure 3.5 shows the extraction workflow. First, the digitisation workflow rasterised the selected floor plan page at 300 dpi to provide sufficient detail for manual tracing, following common document digitisation guidance (National Archives and Records Administration, 2023). The workflow then displayed the rasterised image in an interactive Matplotlib interface (Hunter, 2007). Manual digitisation defined the dwelling and stairwell boundaries within this interface.

The workflow followed the four steps shown in Fig. 3.5. First, it loaded the floor plan into the digitisation interface. The digitisation procedure then traced the dwelling and stairwell polygons manually, scaled the polygons from image coordinates to metric coordinates, and classified the resulting geometries as dwelling or stairwell reference zones. Because the digitisation procedure initially stored the polygons in pixel coordinates, the scaling procedure required a pixel-to-metre conversion factor. Depending on the information available in the source document, the workflow applied one of the three scaling methods listed in Tab. 3.4.

Here, D denotes the rasterisation resolution in dpi, n denotes the scale denominator, L_{real} denotes the known real-world reference length in m, d_{px} denotes the measured image length in pixels, A_{real} denotes the known real-world area in m^2, and A_{px}

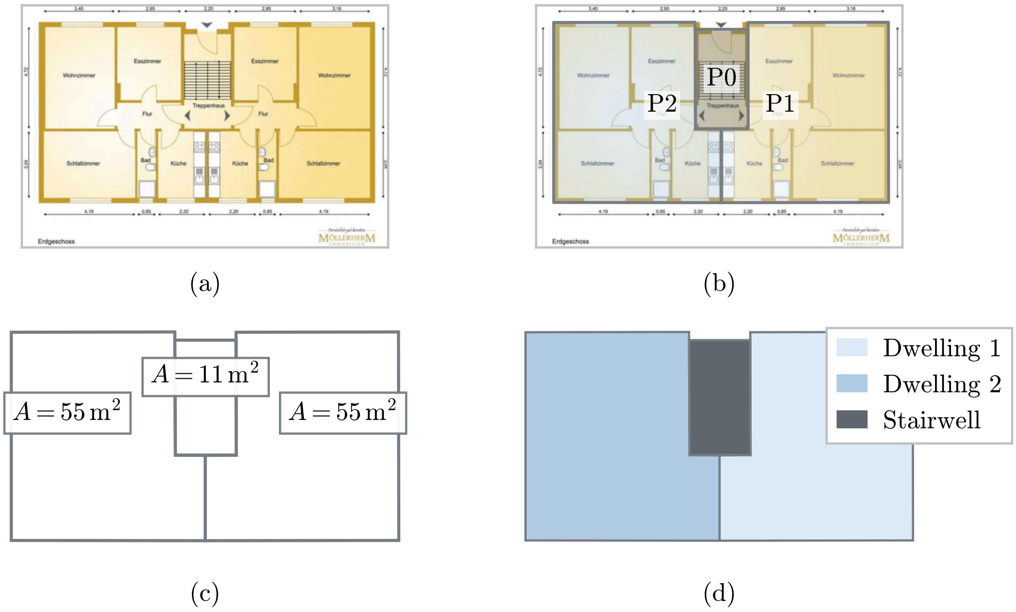


Fig. 3.5: Extraction of reference ground truth from German real estate floor plans. (a) Floor plan displayed in the digitisation interface. (b) Manual digitisation of dwelling and stairwell polygons. (c) Scaling to metric coordinates. (d) Classification into dwelling and stairwell reference zones.

denotes the corresponding digitised polygon area in pixel^2. The resulting conversion factor s has the unit m/pixel.

After selecting the appropriate scaling method, the workflow converted each polygon vertex from image coordinates to metric coordinates according to

$$(x_{\mathrm{m},i}, y_{\mathrm{m},i}) = s(x_{\mathrm{px},i}, y_{\mathrm{px},i}), \tag{3.1}$$

where $(x_{\mathrm{px},i}, y_{\mathrm{px},i})$ denote the image coordinates and $(x_{\mathrm{m},i}, y_{\mathrm{m},i})$ denote the corresponding metric coordinates. The workflow applied the same conversion factor to all polygons within a floor plan to preserve their relative geometry and adjacency. Where the source document provided area information, the consistency check compared the scaled polygon areas with the reported values.

The floor plans in Fig. 3.4 vary in graphical style, drawing quality, level of detail, and available metadata. The selection process retained only clearly interpretable multi-

Tab. 3.4: Scaling methods for converting digitised German real estate floor plans to metric coordinates.

Scaling method	**Available information**	**Pixel-to-metre conversion factor**
Known scale method	Drawing scale, such as 1:50 or 1:100, and rasterisation resolution D	$s_{\text{scale}} = \frac{25.4}{D} \cdot \frac{n}{1000}$
Reference line method	Dimension line or reference element with known real-world length	$s_{\text{ref}} = \frac{L_{\text{real}}}{d_{\text{px}}}$
Area-based method	Known floor area and corresponding digitised polygon area	$s_{\text{area}} = \sqrt{\frac{A_{\text{real}}}{A_{\text{px}}}}$

family residential layouts within the defined validation scope of two to six dwellings and one to two stairwells. The selected plans also required identifiable dwelling and stairwell spaces and usable scale, dimension, or area information. The selection process excluded plans with unclear dwelling boundaries, ambiguous stairwell spaces, insufficient drawing quality, missing or unreliable scale information, or inconsistent interpretations.

3.1.5 Common ground-truth representation

Although the Modified Swiss Dwellings and German real estate branches used different extraction procedures, the two workflows converted both data sources into a common reference ground-truth zoning representation. The output of both branches consisted of:

- the number of dwelling units,
- the geometry of each dwelling unit,
- the number of stairwells,
- the geometry of each stairwell, and
- the spatial arrangement and adjacency of the dwelling and stairwell geometries within the same floor plan coordinate system.

This common representation preserved the individual zone geometries as well as their positions, adjacencies, and overall arrangement within the building footprint. It therefore enabled the subsequent input-preparation workflow to process both datasets consistently and allowed the validation procedure to compare the zones generated by the Multizone Assignment Algorithm with the corresponding reference geometries.

3.1.6 Input preparation for the Multizone Assignment Algorithm

After extracting the reference ground-truth thermal zones, the input-preparation workflow created a separate dataset for the Multizone Assignment Algorithm. This step reproduced the data aggregation and preprocessing stage of the original workflow described in Sec. 2.4.1. In the standard workflow, the required inputs are obtained from external geometric, semantic, and building-stock data. Since these sources were not available for the validation cases, the input-preparation workflow derived a simplified external building footprint by dissolving the internal zone boundaries and generated the remaining equivalent inputs from the extracted reference geometry.

The preparation comprised four steps: generating a simplified external footprint, standardising its orientation with respect to the stairwell or access side, replacing the footprint in an existing CityGML building model, and preparing the auxiliary input assumptions required by the Multizone Assignment Algorithm. These steps enabled the algorithm to process the validation cases without directly using the reference dwelling and stairwell zones as predicted output.

Footprint orientation and CityGML replacement

A simplified footprint was generated for each case by merging the reference dwelling and stairwell polygons and dissolving their internal boundaries. Only this external boundary was retained as algorithm input. Figure 3.6(a)–(b) illustrates this footprint-generation process.

The extracted footprints did not contain address information or an explicitly defined street-facing side, although the Multizone Assignment Algorithm uses this information to estimate the stairwell location. Their orientation was therefore standardised

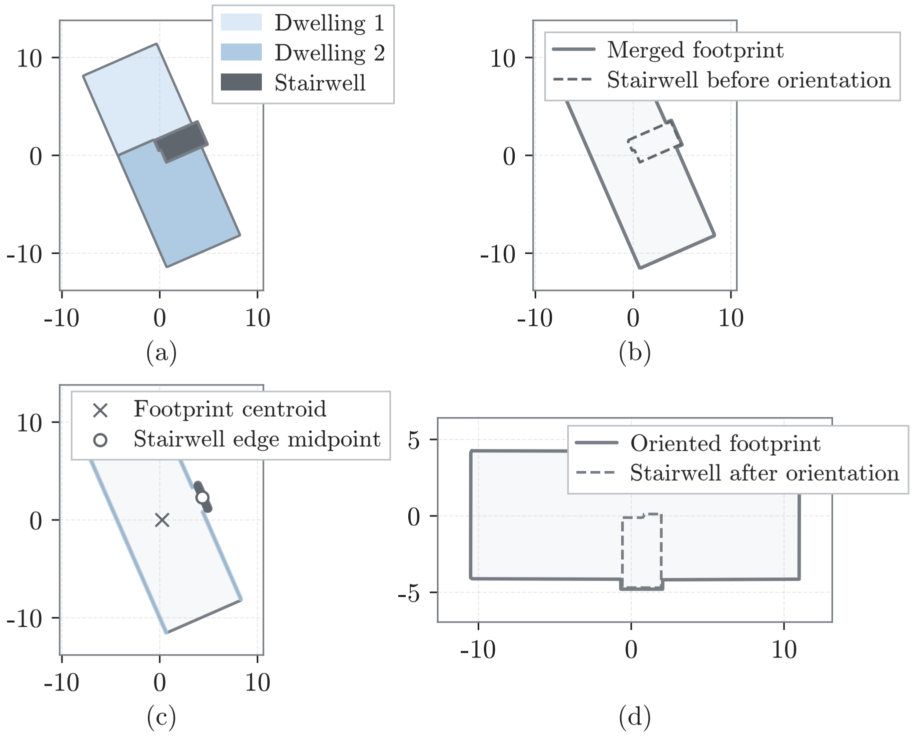


Fig. 3.6: Footprint generation and orientation standardisation for an example building. (a) Reference dwelling and stairwell zones. (b) Simplified external footprint after dissolving the internal zone boundaries. (c) Identification of the stairwell-side façade. (d) Oriented footprint prepared as input for the Multizone Assignment Algorithm.

using the reference stairwell geometry, with the footprint centroid and midpoint of the stairwell-facing boundary defining the likely access direction before rotation and, where necessary, flipping aligned the access façade with the bottom edge, as shown in Fig. 3.6(c)–(d).

To create a CityGML-compatible input for the Multizone Assignment Algorithm, the oriented footprint was inserted into an existing CityGML building template obtained from the Berlin LoD2 building dataset (Senatsverwaltung für Stadtentwicklung, Bauen und Wohnen Berlin, 2024). This replaced the original horizontal geometry while preserving the template's georeferenced position, vertical structure, and address information. Figure 3.7 shows this replacement step.

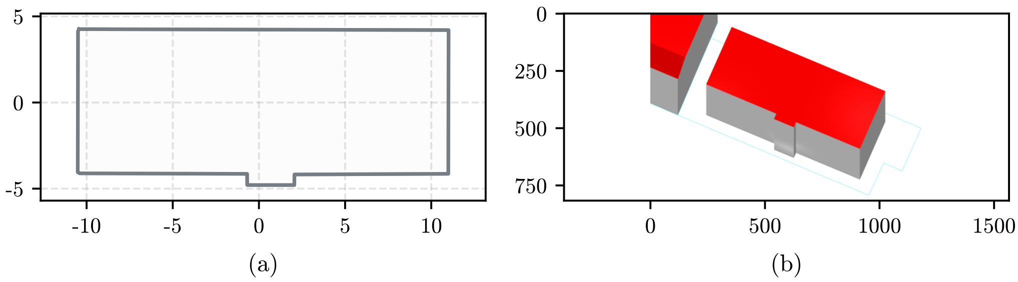


Fig. 3.7: CityGML footprint replacement workflow. (a) Oriented external footprint used as the replacement geometry. (b) CityGML/LoD2 building after footprint replacement while retaining the vertical structure of the template building.

Auxiliary inputs for the zoning run

In addition to the external footprint geometry, the Multizone Assignment Algorithm requires auxiliary information to constrain the expected dwelling subdivision of the building. Depending on the input configuration, the workflow can specify this information either as an estimated mean dwelling area or as an expected number of dwellings. In the standard workflow, address-related data and building-stock assumptions, including construction-year information, provide these inputs. However, the validation cases did not consistently contain this information. The input-preparation workflow therefore prepared the auxiliary inputs separately.

For the Modified Swiss Dwellings cases, this study did not directly use the known dwelling counts or measured dwelling areas of the validation cases as Multizone Assignment Algorithm inputs because doing so would have introduced reference ground-truth information into the automated zoning process. Instead, the study created a separate estimation dataset from Modified Swiss Dwellings buildings outside the validation sample. The estimation models used external building footprint area as the predictor and mean dwelling area as the target variable. The model-selection process compared several simple candidate models, including a constant baseline, a linear regression model, and quantile-based piecewise linear regression models.

Appendix B provides the detailed model comparison, selected regression form, fitted parameters, and final model plot. This estimation step did not form part of the reference ground-truth extraction, and this thesis does not interpret it as a representative dwelling-size model. The study used the estimation model only as a dataset-specific preprocessing step to provide the auxiliary input required to run the Multizone Assignment Algorithm.

Table 3.5 summarises the preparation of the auxiliary inputs for the two validation datasets.

Tab. 3.5: Preparation of auxiliary inputs for the Multizone Assignment Algorithm.

Validation source	Available information	Input preparation	Purpose
German real estate floor plans	Year of construction	Workflow combines the year of construction with the oriented CityGML input	Supports building-stock assumptions.
Modified Swiss Dwellings	No construction year, address, or building-stock metadata	Estimation model derives mean dwelling area from footprint area using non-validation cases	Provides dwelling-size input without using reference dwelling counts or measured dwelling areas.

The input-preparation workflow then prepared the dataset-specific Multizone Assignment Algorithm inputs. For the German real estate cases, the workflow combined the oriented CityGML file with the available year of construction. For the Modified Swiss Dwellings cases, it combined the oriented CityGML file with the estimated mean dwelling area derived from the external building footprint area.

3.1.7 Comparison and validation metrics

For each validation case, the validation framework defined the reference layout as the set of ground-truth dwelling and stairwell polygons G_i, where $i = 1, \ldots, N_{\mathrm{GT}}$. It defined the Multizone Assignment Algorithm output as the set of predicted thermal zone polygons P_j, where $j = 1, \ldots, N_{\mathrm{pred}}$. Here, N_{GT} denotes the number of ground-truth zones, and N_{pred} denotes the number of predicted zones. To evaluate the predicted zones against the reference layout, the matching procedure first matched the ground-truth and predicted polygons geometrically. The metric-calculation procedure then quantified the zone-level deviations.

Before matching, the alignment procedure transformed the predicted and reference geometries into the same local coordinate system. The CityGML-based workflow generated the Multizone Assignment Algorithm output, whereas the floor plan processing workflows stored the reference ground-truth geometries in local floor plan coordinates. The geometric comparison then calculated the intersection over union (IoU) between each ground-truth zone G_i and each predicted zone P_j. The pairwise IoU values formed an overlap matrix. The matching procedure used this matrix to assign predicted zones to ground-truth zones and to quantify how closely the algorithmic zoning reproduced the reference layout after alignment.

The validation workflow applied the Hungarian assignment algorithm to establish one-to-one geometric matches (Kuhn, 1955). The matching procedure converted the pairwise intersection-over-union values into the cost matrix

$$C_{ij} = 1 - \mathrm{IoU}_{ij}, \tag{3.2}$$

so that minimising the total assignment cost maximised the total geometric overlap between matched zones. After the assignment, the matching procedure linked each ground-truth zone G_i to its assigned predicted zone P_{j^*}. It rejected assigned pairs with an IoU below 0.15 as invalid matches. The procedure recorded ground-truth zones without a valid predicted match as false negatives (N_{FN}) and predicted zones without a valid ground-truth match as false positives (N_{FP}).

Based on the matched and unmatched zone information, the metric-calculation procedure computed the metrics listed in Tab. 3.6 to quantify the geometric and topological differences between the reference zoning and the Multizone Assignment Algorithm output.

For the metric definitions, N_{matched} denotes the number of valid matched ground-truth–prediction zone pairs. For each matched pair i, IoU_i, $\mathrm{AreaError}_i$, and d_i denote the intersection-over-union value, relative area error, and centroid distance, respectively. Unmatched ground-truth zones contribute zero to $\mathrm{IoU}_{\mathrm{overall}}$. The sets E_{GT} and E_{pred} denote the dwelling–stairwell adjacency relations in the ground-truth and predicted layouts, respectively.

Tab. 3.6: Metrics used to compare Multizone Assignment Algorithm-generated zoning with ground truth.

Metric	Definition	Interpretation
IoU	$\mathrm{IoU}_{ij} = \dfrac{\mathrm{area}(G_i \cap P_j)}{\mathrm{area}(G_i \cup P_j)}$	Overlap between ground-truth zone G_i and predicted zone P_j.
Mean overall IoU	$\mathrm{IoU}_{\mathrm{overall}} = \dfrac{1}{N_{\mathrm{GT}}} \sum_{i=1}^{N_{\mathrm{GT}}} \mathrm{IoU}_i$	Building-level geometric agreement, including missed ground-truth zones as zero-overlap contributions.
Mean area error in %	$\overline{\mathrm{AreaError}} = \dfrac{1}{N_{\mathrm{matched}}} \sum_{i=1}^{N_{\mathrm{matched}}} \mathrm{AreaError}_i$	Building-level mean relative area difference across all valid matched zone pairs.
Mean centroid distance in m	$\overline{d} = \dfrac{1}{N_{\mathrm{matched}}} \sum_{i=1}^{N_{\mathrm{matched}}} d_i$	Building-level mean spatial displacement across all valid matched zone pairs.
Zone-count deviation	$\Delta N = N_{\mathrm{pred}} - N_{\mathrm{GT}}$	Difference between predicted and ground-truth zone counts.
Topology error	$E_{\mathrm{missing}} = E_{\mathrm{GT}} \setminus E_{\mathrm{pred}}$, $E_{\mathrm{extra}} = E_{\mathrm{pred}} \setminus E_{\mathrm{GT}}$	Missing or additional dwelling–stairwell adjacency relations after zone matching.

The comparison metrics and matching indicators identify structural, geometric, and topological deviations in the thermal zoning generated by the algorithm. The matching indicators N_{FN} and N_{FP} identify missing ground-truth zones and additional predicted zones, respectively. The zone-count deviation indicates whether the algorithm generated too few or too many thermal zones.

The IoU, mean area error, and mean centroid distance describe differences in zone shape, size, and spatial position. The topology error evaluates whether the predicted zoning preserves the dwelling–stairwell adjacency relations of the reference layout. These deviations are relevant to the subsequent zoning-variant simulation study because changes in the number, size, position, and adjacency of zones can alter façade allocation, internal gains, thermal capacity, solar exposure, and interzonal heat exchange. The validation therefore identifies the types of zoning deviations that the automated workflow produces, while the subsequent simulation study evaluates how comparable zoning differences affect annual heating demand, peak heating load, and overheating hours.

3.1.8 Diagnostic classification

The diagnostic framework used the calculated zone-level and building-level metrics to assign one diagnostic outcome to each validation case. The classification did not replace the individual metrics but translated them into an interpretable case-level result. While the metrics quantify separate aspects of the zoning difference, the diagnostic outcome identifies the dominant deviation for each building. It indicates whether the algorithmic zoning is structurally usable, whether it preserves the dwelling–stairwell topology, and which geometric deviation is most relevant for interpreting the case.

The classification procedure applied the rules in Tab. 3.7 sequentially from top to bottom and assigned each case to the first matching category. It first checked for structural problems, then for topology problems, and finally for geometric deviations. The diagnostic classification used study-specific thresholds for the geometric and topological metrics. The applied thresholds and the corresponding classification rules are also summarised in Tab. 3.7.

Tab. 3.7: Hierarchical diagnostic rules for classifying the validation cases.

Diagnostic category	**Evaluation stage**	**Additional rule applied at this stage**
Structural mismatch	Structural check	$N_{\mathrm{FN}} > 0$ and $N_{\mathrm{FP}} > 0$
Under-zoned	Structural check	$N_{\mathrm{FN}} > 0$ or $\Delta N < 0$
Over-zoned	Structural check	$N_{\mathrm{FP}} > 0$ or $\Delta N > 0$
Correct count but topology mismatch	Topology check	$\lvert E_{\mathrm{missing}} \rvert > 0$ or $\lvert E_{\mathrm{extra}} \rvert > 0$
Weak geometric match	Geometric check	$\mathrm{IoU}_{\mathrm{overall}} < 0.50$
Correct topology but high area error or shifted zones	Geometric check	$\overline{\mathrm{AreaError}} > 40\,\%$ or $\overline{d} > 3.0\,\mathrm{m}$
Good geometric and topological match	Geometric check	$\mathrm{IoU}_{\mathrm{overall}} \geq 0.70$, $\overline{\mathrm{AreaError}} \leq 40\,\%$, and $\overline{d} \leq 3.0\,\mathrm{m}$
Moderate geometric match	Geometric check	$0.50 \leq IoU_{\mathrm{overall}} < 0.70$, $\mathrm{AreaError} \leq 40\,\%$, and $d \leq 3.0\,\mathrm{m}$

The qualitative review used selected ground-truth–prediction overlay figures to support the interpretation of the diagnostic results. It focused mainly on cases with diagnostic deviations. The overlays linked the numerical diagnosis to the spatial layout and helped identify deviation mechanisms such as merged dwellings, split dwelling regions, displaced stairwells, incorrectly allocated areas, and missing dwelling–stairwell adjacencies.

The final validation interpretation combined the building-level diagnostic table, the distribution of diagnostic categories, and representative spatial overlays. After defining these evaluation rules, the following subsection describes the implementation and storage structure of the validation workflow to ensure traceability and reproducibility.

3.1.9 Implementation setup

This study implemented the validation workflow as a reproducible Python-based processing pipeline. The pipeline assigned a distinct identifier to each validation case, which allowed consistent tracking of the reference floor plan, the prepared Multizone Assignment Algorithm input, the algorithmic zoning output, the comparison metrics, the diagnostic classification, and the overlay figures throughout the workflow.

Table 3.8 summarises the principal intermediate and output files. The file-based structure separated the main processing stages and supported reproducible validation results. The stored ground-truth files provided a fixed reference state, while the generated-zone files contained the Multizone Assignment Algorithm output before alignment, matching, and metric calculation.

For the Modified Swiss Dwellings cases, the implementation started from graph-based room geometries and connectivity data. For the German real estate cases, it started from manually digitised and scaled dwelling and stairwell representations. After ground-truth extraction, the two processing branches converted both datasets into the same polygon-based dwelling–stairwell representation. The pipeline then processed them consistently through input preparation, automatic zoning, alignment, matching, metric calculation, and diagnostic classification.

The pipeline generated zone-level outputs, building-level metric summaries, diagnostic tables, and overlay figures for visual analysis. This study used these outputs to support the quantitative validation results and the qualitative analysis of representative cases. The implementation and evaluation scripts for the validation pipeline are available in an accompanying GitHub repository (Thomas, 2026).

Tab. 3.8: Principal intermediate and output files in the validation pipeline.

Pipeline stage	Format	Purpose
Reference ground truth	`.pkl`	Stores dwelling and stairwell polygons, zone labels, and case identifiers.
Simplified footprint transfer	`.geojson`	Transfers the oriented footprint geometry to the CityGML input-preparation stage.
Multizone Assignment Algorithm input	CityGML / Level of Detail 2	Provides the algorithm-compatible building geometry after footprint replacement.
Generated zoning output	`.pkl`	Stores the dwelling and stairwell zones generated by the Multizone Assignment Algorithm.
Metrics and diagnostic results	`.csv`	Stores zone-level metrics, building-level metrics, and diagnostic classifications.
Overlay figures	`.pdf` / `.png`	Support visual comparison of the ground-truth and algorithm-generated zones.

3.2 Zoning-variant sensitivity analysis

The zoning-variant sensitivity analysis builds on the geometric validation described in Sec. 3.1. The validation evaluates whether the Multizone Assignment Algorithm reproduces the reference dwelling–stairwell zoning structure from limited inputs. However, geometric validation alone does not indicate whether zoning deviations affect building energy simulation results. A zoning deviation may be geometrically visible, but its effect on annual heating demand, peak heating load, or overheating hours depends on how it changes the thermal zone structure, interzonal heat exchange, operational assumptions, and stairwell treatment.

The sensitivity analysis therefore uses controlled zoning variants to evaluate how simulation outputs respond to different zoning assumptions. These variants do not reproduce individual validation cases directly. Instead, they represent the main zoning-deviation mechanisms that the validation identified, including aggregation, reduced or increased zoning granularity, alternative internal boundary placement, omission of the stairwell zone, and changes in the thermal treatment of the stairwell.

A comparison based on only one fixed input configuration would show the effect of the zoning variants only for that specific set of assumptions. However, the Multizone Assignment Algorithm-based workflow targets modelling contexts with limited information, where both the zoning representation and several physical and operational inputs remain uncertain.

The zoning-variant sensitivity analysis therefore evaluates whether differences between the zoning variants remain relevant for annual heating demand, peak heating load, and overheating hours when the physical and stochastic simulation inputs also vary. It thus forms the simulation-based continuation of the validation framework: the validation assesses how accurately the algorithm-generated zoning reproduces the reference zoning, whereas the zoning-variant sensitivity analysis assesses the consequences of zoning differences for the selected energy-performance indicators.

3.2.1 Simulation setup

The zoning-variant simulations were implemented through a Python-based workflow. For each simulation run, TEASER generated a reduced-order Modelica building model using components from AixLib. The workflow then applied the sampled construction, setpoint, window, internal-gain, occupancy, and weather assumptions to the generated model.

The Modelica models were compiled and simulated in Dymola (Dassault Systèmes, 2026) through Python. Python scripts controlled model generation, parameter assignment, simulation execution, and the extraction of the run-level performance indicators. Each simulation covered one full year with hourly input profiles.

The simulation workflow assigned construction-related thermal properties through the TABULA building typology (Loga et al., 2016). It treated heating setpoints as operational control assumptions, applied the window-to-wall ratio factor as a façade and window-area modifier, and represented occupant- and appliance-related internal gains using stochastic household profiles generated with the LoadProfileGenerator (Pflugradt et al., 2022).

The multi-zone variants retained interzonal heat transfer because they differed in zone count, adjacency relations, and stairwell treatment. These differences can affect heat exchange between dwellings, neighbouring zones, and the stairwell and may therefore influence the performance indicators.

Figure 3.8 shows the simulation workflow for the sensitivity analysis. The simulation framework combined controlled zoning variants, sampled uncertain input configurations, and stochastic occupancy profiles in annual TEASER–AixLib simulations. The post-processing procedure then prepared the simulation results for the comparison of the zoning variants.

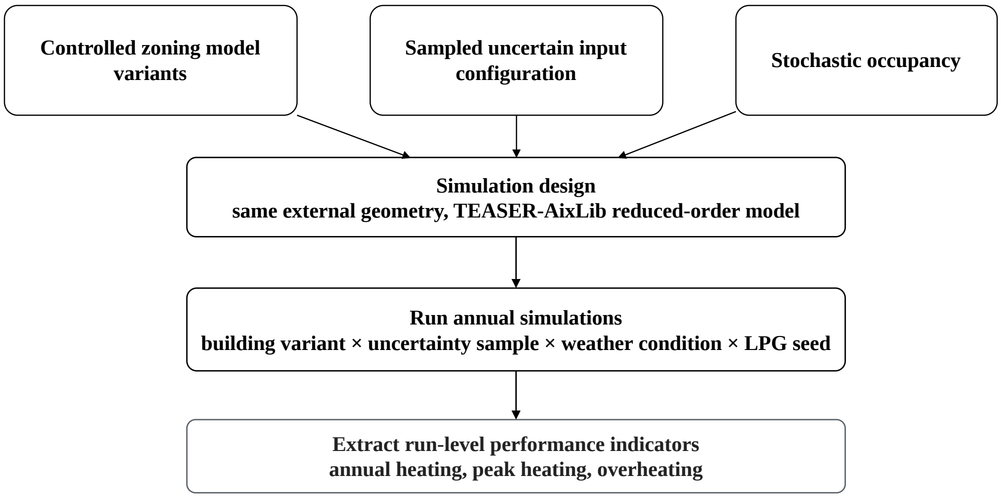


Fig. 3.8: Workflow from zoning variants, uncertain input configurations, and stochastic occupancy profiles to annual TEASER–AixLib simulations and performance indicators.

Each run was identified by its zoning variant, uncertainty-sample identifier, weather condition, and stochastic occupancy seed. The post-processing scripts aggregated the results by zoning variant, weather condition, and construction state. They calculated the median values and percentile ranges used for the summary tables and comparison plots presented in Sec. 4.2.

For the zoning-variant comparison, all zoning variants used the same external building geometry. The sampled boundary conditions and uncertain inputs are described in Sec. 3.2.3. The variants therefore differed mainly in their internal thermal zoning and stairwell representation. This setup enabled the analysis to interpret differences in annual heating demand, peak heating load, and overheating hours as responses to the zoning structure and stairwell treatment while accounting for variations in the remaining simulation inputs.

Space cooling was disabled, and no ideal cooler was used. Overheating hours therefore represent temperature exceedance without active cooling. They were calculated as the annual number of hours during which at least one thermal zone exceeded 26 °C. The threshold was selected as a simple comparative limit because 26 °C is used in residential overheating guidance (Chartered Institution of Building Services Engineers, 2017).

As a supplementary assessment, computational effort was evaluated for all 60 samples per zoning variant. The workflow recorded the simulation runtime for each completed run, and the median runtime was calculated separately for each zoning variant. The median was used to represent the typical runtime while reducing the influence of unusually long individual runs. This assessment provides additional context for the practical scalability of the zoning representations.

3.2.2 Zoning model variants

This study defined the zoning variants to evaluate whether the zoning deviations that the geometric validation identified could influence simulation outputs. The variants did not reproduce individual validation cases directly. Instead, they provided controlled analogues of the main zoning mechanisms observed in the validation results, including aggregation, finer subdivision, alternative internal boundary placement, missing stairwell representation, and changed stairwell treatment. The simulation case represented a five-storey multi-family residential building.

Table 3.9 summarises the variant definitions, and Fig. 3.9 shows their schematic layout arrangements.

For variants with an explicit stairwell representation, the simulation model represented the stairwell or circulation area as a separate thermal zone. The standard multi-zone variants treated the stairwell as unheated, while V8 treated the stairwell as a heated zone with a fixed setpoint of 21 °C. This distinction is relevant because

Tab. 3.9: Zoning variants and their relation to diagnostic deviation mechanisms.

Variant	Residential zoning	Stairwell	State	Zones	Related diagnostic deviation
V1	Aggregated building	No	–	1	Strong under-zoning or structural simplification
V2	One zone per storey	Yes	Unheated	6	Under-zoning through storey-level aggregation
V3	Two zones per storey, Layout A	Yes	Unheated	11	Reference multi-zone comparison case
V4	Two zones per storey, Layout B	Yes	Unheated	11	Shifted boundaries or area redistribution at fixed zone count
V5	Four zones per storey, Layout A	Yes	Unheated	21	Over-zoning through finer subdivision
V6	Four zones per storey, Layout B	Yes	Unheated	21	Over-zoning with shifted boundaries or area redistribution
V7	Two zones per storey, Layout A	No	–	10	Missing explicit stairwell representation or topology simplification
V8	Two zones per storey, Layout A	Yes	Heated	11	Changed stairwell thermal treatment

the stairwell can act as an intermediate thermal zone between dwellings and may influence interzonal heat exchange, heating demand, and peak heating load.

The variants covered different zoning granularities, ranging from a fully aggregated single-zone model to storey-level, dwelling-level, and finer residential subdivisions. The variant set also included configurations with the same number of zones but different layout arrangements to test whether internal boundary placement, area redistribution, and adjacency relations affected the simulation outputs.

The variants did not reproduce individual validation layouts but isolated selected zoning mechanisms under controlled simulation conditions. Instead, they isolated selected zoning mechanisms under controlled simulation conditions. The relevant simulation effects therefore relate to the number of zones, the spatial arrangement of internal boundaries, adjacency relations, separating surfaces, and the thermal treatment of the stairwell.

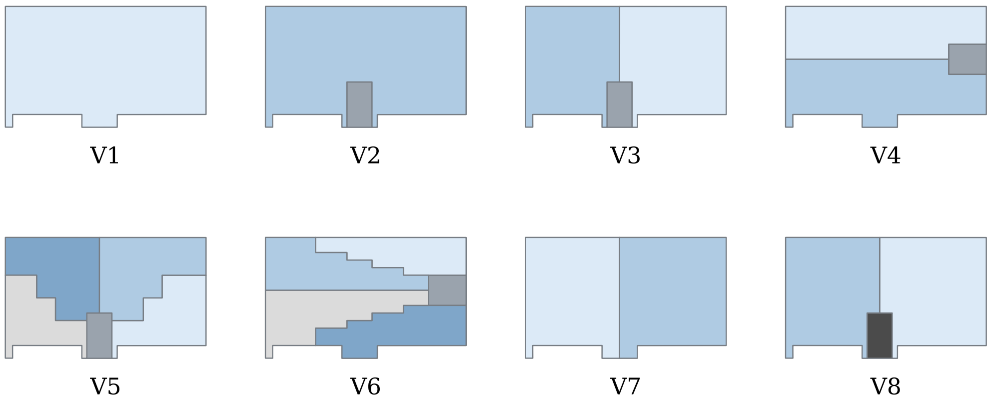


Fig. 3.9: Schematic layouts of zoning variants V1–V8 in the sensitivity analysis.

3.2.3 Boundary conditions and input uncertainties

The zoning-variant sensitivity analysis sampled uncertain input assumptions required for thermal enrichment and simulation. These inputs represent information that is frequently unavailable, simplified, or estimated in limited-information urban building-modelling workflows. They defined the range of conditions under which the zoning variants were compared. The inputs were sampled jointly, and the zoning variants described in Sec. 3.2.2 were simulated across the resulting conditions to assess whether zoning-related differences remained observable despite input uncertainty.

Table 3.10 summarises the sampled inputs and fixed boundary-condition choices. The upper part of the table describes the general simulation design, while the lower part lists the uncertain physical and modelling inputs included in the sensitivity analysis.

The TABULA construction-age class and retrofit state represented typological enrichment uncertainty. In the TEASER workflow, these inputs determined the construction-related thermal properties that the TABULA building typology assigned (Loga et al., 2016). Since LoD2 geometry does not provide reliable construction-period information and European building-stock data can be incomplete or inconsistent (Castellazzi et al., 2016), the sampling procedure either retained the original TABULA class or shifted it to a neighbouring class. This study based the prob-

Tab. 3.10: Boundary conditions and uncertain inputs in the zoning-variant sensitivity analysis.

Category	Setting or range
General simulation design	
Zoning variants	V1–V8
Sampling structure	Building variant × uncertainty sample × weather condition × stochastic occupancy seed.
Uncertainty samples	60 samples for each zoning variant.
Occupancy seeds	Four stochastic occupancy seeds for each uncertainty sample.
Simulation period	One full year.
Uncertain physical and modelling inputs	
Construction state	Sampling assigns the standard state with $p_{\text{standard}} = 0.75$ and the retrofit state otherwise.
TABULA age class	Sampling retains the original TABULA age class or shifts it to a neighbouring class by -1, 0, or $+1$ class.
Heating setpoint	Mean residential heating setpoint: 20.5 to 21.5 °C.
Zonal setpoint spread	0.5 to 2.0 K spread between residential zones in the multi-zone variants.
Residential setpoint	Zone-level heating setpoints limited to 18 to 24 °C.
Window-to-wall ratio	Window-to-wall ratio scaling factor: 0.8 to 1.2.
Internal gains	Internal-gain scaling factor: 0.8 to 1.2.
Weather	Two annual weather cases: Napoli 2011 and Munich 2015.

ability of retaining the original class on the residential building-age classification accuracy reported by Rosser et al. (2019). It used the retrofit-state probability $p_{\text{standard}} = 0.75$ as a proxy for the share of non-retrofitted or energy-inefficient buildings in the European building stock (European Commission, 2020).

The heating setpoint and zonal setpoint spread represented operational uncertainty. The sampling procedure first selected a building-level mean heating setpoint and then applied an additional zonal spread to the multi-zone variants. It limited the resulting zone-level heating setpoints to 18 to 24 °C, following literature-supported residential heating bounds and indoor-temperature guidance (Kull et al., 2020; Public Health England, 2014). The zonal spread represented heterogeneous dwelling temperature preferences and remained relevant because temperature differences between dwellings and the stairwell can drive interzonal heat exchange. The single-zone variant could not represent this heterogeneity explicitly and therefore used a uniform building-level setpoint.

The LoadProfileGenerator generated stochastic household profiles for occupancy and appliance use (Pflugradt et al., 2022). The profiles represented one- to five-person households and were assigned to the residential zones using four stochastic occupancy seeds for each uncertainty sample. In the dwelling-resolved variants, each dwelling zone received one household profile. The workflow therefore used ten profiles for variants with two dwelling zones per storey and twenty profiles for variants with four dwelling zones per storey.

For the coarser zoning variants, the same underlying household profiles were combined within the aggregated residential zones. Occupant and appliance gains were summed so that aggregation changed their spatial distribution rather than introducing a different set of households. The stairwell received no residential household profile.

The sensitivity analysis treated the window-to-wall ratio factor as a façade and window-area modifier. This factor represented uncertainty in opening geometry and façade assumptions when detailed window information was unavailable. The implementation followed the concept of Energy Design Measures, which vary façade- and window-related properties parametrically (Hale et al., 2008). The selected window-to-wall ratio range of 0.8 to 1.2 corresponds to a ±20 % variation around the baseline and follows previous work in which Rumnici & Abualdenien (2019) varied the window-to-wall ratio by the same order of magnitude.

The simulation workflow additionally scaled internal gains by a factor of 0.8 to 1.2, corresponding to a ±20 % variation around the baseline. This range is consistent with the internal-gain scaling factor reported by Brès et al. (2022). The zoning-variant sensitivity analysis also included weather as an external boundary-condition uncertainty. The simulations used a Napoli–Capodichino weather file obtained from Climate.OneBuilding and a Munich TRY2015 weather file obtained from the German Weather Service to represent warm and cold climatic conditions and test whether zoning-related effects remained consistent across the two weather cases (Climate.OneBuilding.Org, 2026; Deutscher Wetterdienst, 2026).

4 Results

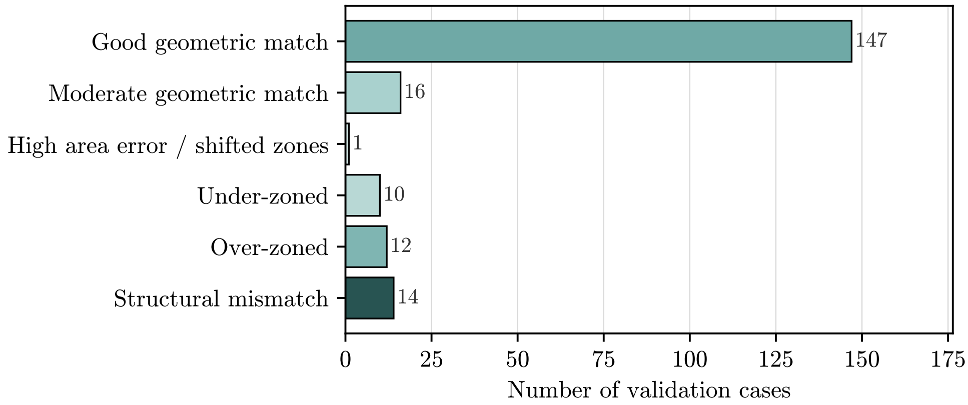
This chapter presents the outcomes of the validation and sensitivity analysis that the research method chapter describes. It reports the results descriptively and focuses on the calculated metrics, diagnostic classifications, and observed distributions before the discussion chapter interprets their implications.

4.1 Validation results

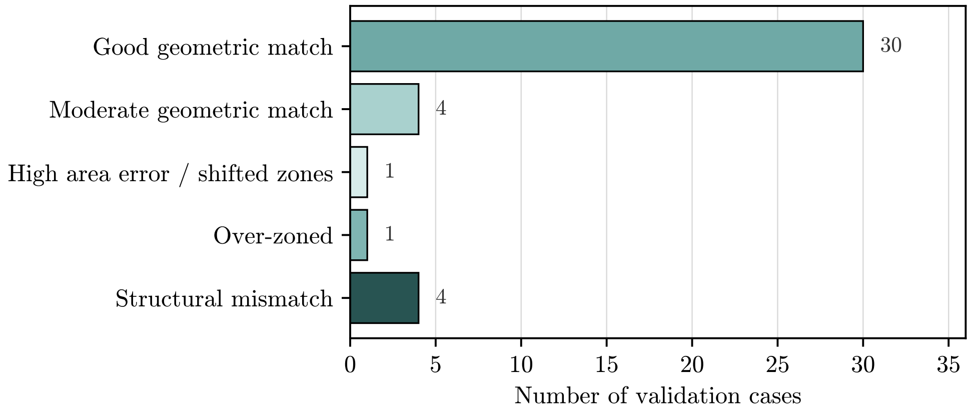
This section reports the geometric and topological agreement between the dwelling and stairwell zones that the Multizone Assignment Algorithm generates and the corresponding reference ground-truth zones. The analysis considers the Modified Swiss Dwellings dataset and the German real estate dataset separately because the two datasets differ in their source formats and ground-truth extraction procedures. The diagnostic classifications summarise the validation outcomes across all cases and support a direct comparison between the two datasets.

The diagnostic categories were assigned using the hierarchical rules defined in Tab. 3.7. A good match preserves the reference zone count and dwelling–stairwell topology and satisfies $IoU_{\text{overall}} \geq 0.70$, AreaError $\leq 40\,\%$, and $d \leq 3.0\,\text{m}$. A moderate match also preserves the zone count and topology and remains within the area-error and centroid-distance limits, but has an overall IoU between 0.50 and 0.70.

Figure 4.1 compares the diagnostic classifications of the two validation datasets. In both datasets, the good geometric and topological match category contains the largest number of cases.



(a) Modified Swiss Dwellings validation cases.



(b) German real estate validation cases.

Fig. 4.1: Diagnostic classification of the Modified Swiss Dwellings and German real estate validation cases. Bars show the number of cases in each diagnostic category.

For the Modified Swiss Dwellings dataset, the classification assigns 147 of the 200 cases to the good geometric and topological match category and 16 cases to the moderate geometric match category. The remaining cases comprise 10 under-zoned cases, 12 over-zoned cases, 14 structural mismatches, and one case with high area error or shifted zones. Zone-count differences and structural mismatches therefore occur more frequently than isolated differences in zone area or position.

For the German real estate dataset, the classification assigns 30 of the 40 cases to the good geometric and topological match category and four cases to the moderate geometric match category. The remaining cases comprise one over-zoned case, one case with high area error or shifted zones, and four structural mismatches.

The high-area-error or shifted-zone case in each dataset retains the reference zone count and dwelling–stairwell topology but exceeds the defined threshold for relative area error or centroid displacement.

Tab. 4.1: Aggregated validation outcomes for the two validation datasets.

Outcome	Modified Swiss Dwellings	German real estate
Good geometric match	147 (73.5 %)	30 (75.0 %)
Moderate geometric match	16 (8.0 %)	4 (10.0 %)
Deviation-related cases	37 (18.5 %)	6 (15.0 %)
Total	200 (100 %)	40 (100 %)

The deviation-related group combines structural mismatch, under-zoned, over-zoned, and high-area-error or shifted-zone cases.

Table 4.1 shows that good and moderate outcomes together account for 81.5 % of the Modified Swiss Dwellings cases and 85.0 % of the German real estate cases. Deviation-related cases account for the remaining 18.5 % and 15.0 %, respectively.

4.1.1 Representative validation cases

The following qualitative comparison examines characteristic examples from the diagnostic classification. The selected cases cover good geometric agreement and the principal deviation-related outcomes. The good-match examples represent cases in which the Multizone Assignment Algorithm reproduces the intended dwelling–stairwell structure with only minor local boundary differences. The remaining examples clarify how structural mismatches, zone-count differences, and deviations in zone area or position appear in the geometric comparison.

Modified Swiss Dwellings validation cases

In the Modified Swiss Dwellings good-match cases in Fig. 4.2, the algorithm-generated dwelling outlines closely follow the reference dwelling geometries, and the algorithm positions the stairwell close to the reference stairwell. Small deviations remain along some internal boundaries and stairwell edges, but the algorithm reproduces the overall dwelling–stairwell arrangement. These examples correspond to the high proportion of good geometric and topological matches in the diagnostic classification.

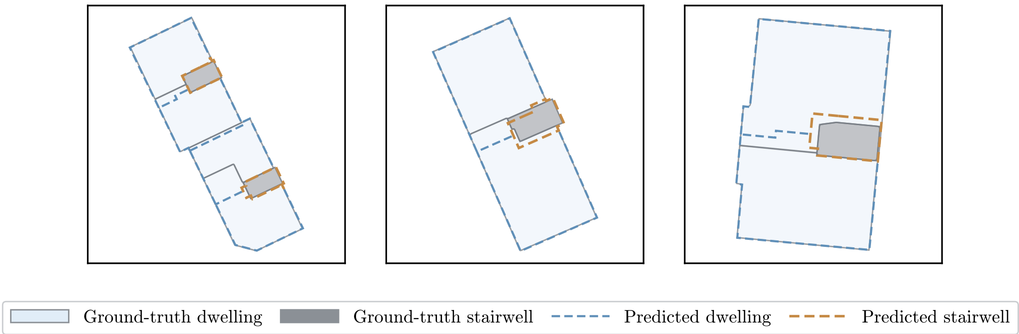


Fig. 4.2: Representative Modified Swiss Dwellings validation cases with good geometric and topological agreement. Solid fills represent the reference ground-truth zones, while dashed outlines represent the zones generated by the Multizone Assignment Algorithm.

Fig. 4.3: Representative Modified Swiss Dwellings validation cases with diagnostic deviations. Solid fills represent the reference ground-truth zones, while dashed outlines represent the zones generated by the Multizone Assignment Algorithm.

The Modified Swiss Dwellings cases in Fig. 4.3 represent different deviation-related outcomes. In the structural mismatch examples, the algorithm-generated dwelling–stairwell arrangement differs from the reference topology. The over-zoned case contains more subdivisions than the reference structure, whereas the under-zoned case contains fewer subdivisions.

The high-area-error or shifted-zone case differs from these structural and zone-count deviations. The generated zoning retains the general dwelling–stairwell arrangement and the reference number of zones, but one or more corresponding zones differ substantially in area or centroid position. The overlay therefore indicates a geometric allocation error rather than a change in the basic zoning structure.

German real estate validation cases

Comparable geometric agreement appears in the German real estate examples in Fig. 4.4. Despite the different source format and manual ground-truth extraction procedure, the algorithm-generated zones reproduce the main dwelling–stairwell arrangement with only minor local boundary differences. The close alignment between the generated zone outlines and the reference geometries provides a qualitative counterpart to the high number of good geometric and topological matches in this dataset.

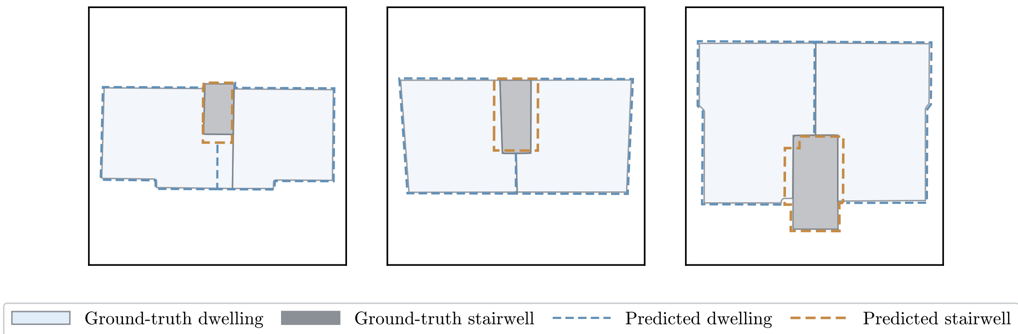


Fig. 4.4: Representative German real estate validation cases with good geometric and topological agreement. Solid fills represent the reference ground-truth zones, while dashed outlines represent the zones generated by the Multizone Assignment Algorithm.

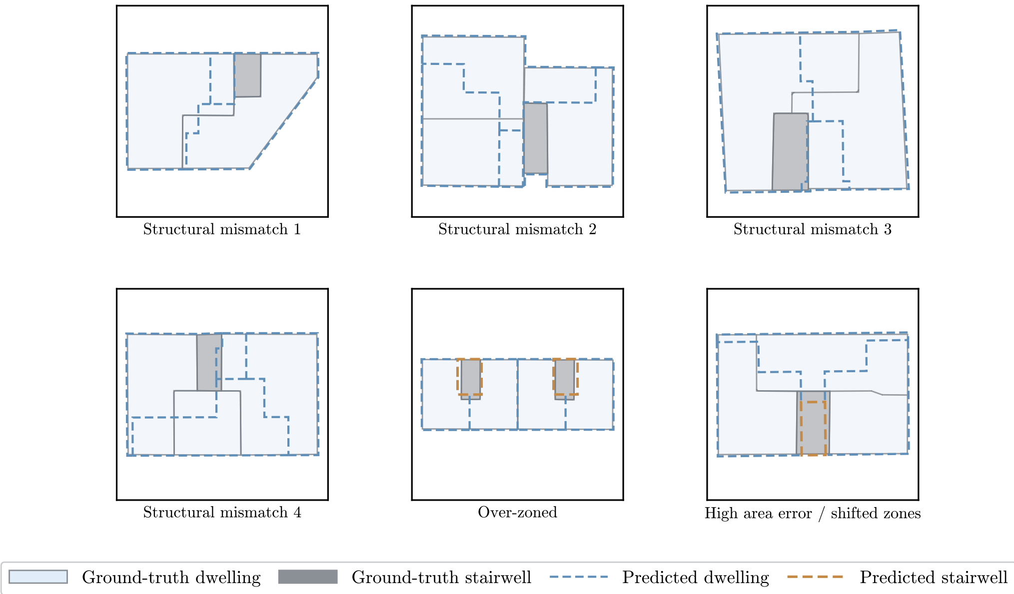
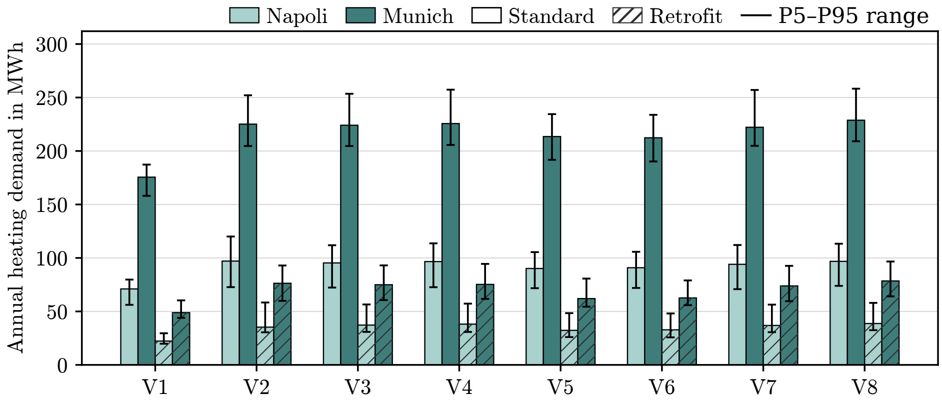


Fig. 4.5: Representative German real estate validation cases with diagnostic deviations. Solid fills represent the reference ground-truth zones, while dashed outlines represent the zones generated by the Multizone Assignment Algorithm.

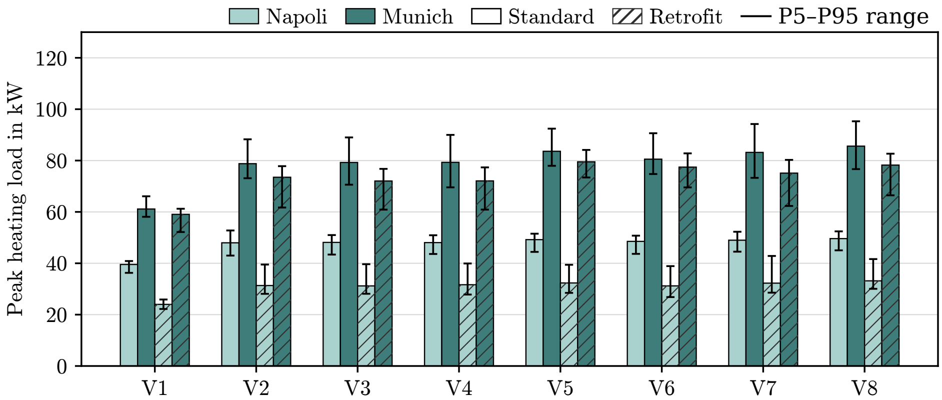
Among the German real estate examples in Fig. 4.5, the structural mismatches alter the dwelling–stairwell topology, whereas the over-zoned case introduces an additional subdivision. The high-area-error or shifted-zone case preserves the general arrangement and the same zone count as the reference layout but differs substantially in the area or centroid position of one or more matched zones. This case therefore represents a spatial allocation error rather than a change in zone count or dwelling–stairwell topology.

4.2 Results of the zoning-variant sensitivity analysis

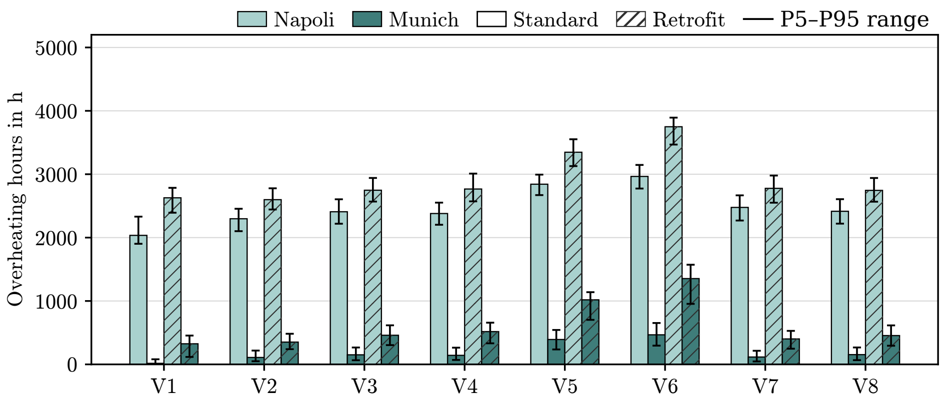
The zoning-variant sensitivity analysis compares variants V1–V8 for the standard and retrofit construction states under the two weather conditions. The median values of the performance indicators are summarised in Tab. 4.2. The corresponding visual comparison, including the 5th–95th percentile uncertainty ranges, is provided in Fig. 4.6. Detailed median and percentile values are reported in Appendix C.



(a) Annual heating demand.



(b) Peak heating load.



(c) Overheating hours.

Fig. 4.6: Comparison of zoning variants V1–V8 across annual heating demand, peak heating load, and overheating hours for the Napoli and Munich weather conditions and the standard and retrofit construction states. Error bars show the 5th–95th percentile ranges.

Tab. 4.2: Median performance-indicator values for the zoning variants under the standard and retrofit construction states.

Variant	Annual heating demand in MWh		Peak heating load in kW		Overheating hours in h	
	Napoli	**Munich**	**Napoli**	**Munich**	**Napoli**	**Munich**
Standard construction state						
V1	71.7	175.2	39.5	61.1	2034	16
V2	96.9	224.9	48.2	78.8	1867	23
V3	95.7	223.8	47.8	79.2	1882	16
V4	97.0	225.5	47.8	79.3	1875	10
V5	90.5	213.2	49.2	83.6	1989	24
V6	90.6	211.7	48.8	80.7	1963	11
V7	93.7	222.2	48.8	83.2	1923	10
V8	97.2	228.8	49.2	85.5	1882	16
Retrofit construction state						
V1	22.7	49.0	23.7	59.1	2624	330
V2	35.8	76.2	30.7	73.5	2313	211
V3	37.6	75.5	31.1	72.1	2332	213
V4	38.6	76.5	31.3	72.2	2276	163
V5	32.3	62.2	32.3	79.6	2439	242
V6	33.1	63.2	31.3	77.4	2378	203
V7	37.3	74.3	32.2	75.2	2346	161
V8	39.0	79.0	33.0	78.3	2329	210

Across all combinations of weather and construction state, the median values in Tab. 4.2 identify V1 as the variant with the lowest annual heating demand and peak heating load. In the standard construction state, annual heating demand increases from 71.7 MWh for V1 to 90.5 to 97.2 MWh for the multi-zone variants under Napoli conditions. Under Munich conditions, it increases from 175.2 MWh for V1 to 211.7 to 228.8 MWh for the multi-zone variants. In the retrofit construction state, annual heating demand increases from 22.7 MWh for V1 to 32.3 to 39.0 MWh for the multi-zone variants under Napoli conditions and from 49.0 MWh to 62.2 to 79.0 MWh under Munich conditions.

Peak heating load follows a similar ordering. In the standard construction state, V1 has median peak heating loads of 39.5 kW under Napoli conditions and 61.1 kW under Munich conditions. The multi-zone variants range from 47.8 to 49.2 kW under Napoli conditions and from 78.8 to 85.5 kW under Munich conditions. In the retrofit construction state, V1 again has lower median peak heating loads than the multi-zone variants, with 23.7 kW under Napoli conditions and 59.1 kW under Munich conditions. Within the multi-zone group, V2 remains close to V3 and V4, while V5, V6, V7, and V8 have higher peak heating load values in several cases.

Overheating hours follow a different ordering from the heating-related indicators. Under Napoli conditions, V1 has 2034 h in the standard construction state and 2624 h in the retrofit construction state. The multi-zone variants range from 1867 to 1989 h in the standard state and from 2276 to 2439 h in the retrofit state. Under Munich conditions, the standard construction state produces substantially fewer overheating hours, with values ranging from 10 to 24 h across the variants. In the retrofit construction state, the values range from 161 to 330 h. The finer subdivision variants V5 and V6 have more overheating hours than V3 and V4 in several cases, while V7 and V8 remain close to the corresponding two-zone-per-storey variants.

The plotted values in Fig. 4.6 confirm the median patterns reported in Tab. 4.2 and additionally highlight the 5th–95th percentile uncertainty ranges. The error bars of several multi-zone variants overlap within the same construction state and weather condition. For example, in the annual heating-demand panel under standard Munich conditions, the uncertainty ranges of V2–V4 span approximately 205 to 257 MWh, while those of V5–V6 span approximately 190 to 235 MWh. The peak heating load ranges cover a narrower interval in absolute units. Under standard Napoli conditions, the multi-zone variants concentrate at approximately 43 to 53 kW, while their standard Munich ranges extend to approximately 70 to 95 kW. In the overheating hours panel, the standard Munich case has low median values but visible upper uncertainty limits for several variants. For example, V3 spans approximately 6 to 52 h, while V5 spans approximately 7 to 64 h. The Napoli cases show larger absolute ranges and stronger overlap among the multi-zone variants.

Overall, the largest differences separate the fully aggregated single-zone model V1 from the multi-zone variants. Differences within the multi-zone group are smaller and overlap with the propagated 5th–95th percentile uncertainty ranges in several cases. Annual heating demand and peak heating load follow this pattern, whereas overheating hours show a different ordering. V1 generally produces more overheating hours than the multi-zone variants or values close to the maximum across the variants.

A clear difference between V1 and the multi-zone variants appears during the selected peak heating event window in Fig. 4.7. The vertical dashed line marks the time of peak heating load, at which V1 has the lowest value and the multi-zone variants produce higher loads.

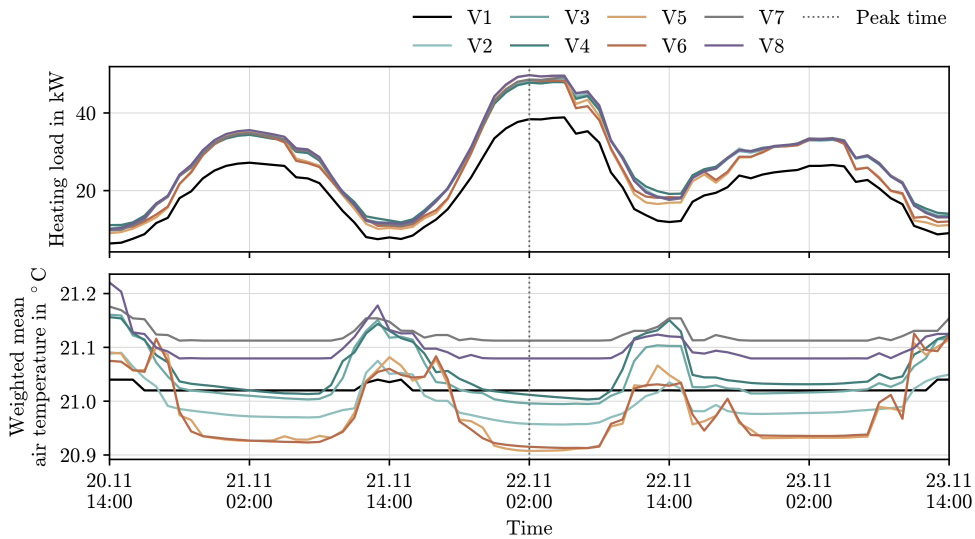


Fig. 4.7: Heating load and weighted mean air temperature during a selected peak heating event for the zoning variants. The upper panel shows the heating load, and the lower panel shows the corresponding weighted mean air temperature. The vertical dashed line marks the time of peak heating load.

Across the zoning variants, the selected three-day overheating event in Fig. 4.8 follows a similar diurnal pattern. Weighted mean air temperatures increase during the daytime and decrease during the night and early morning, while the dashed horizontal line marks the overheating threshold of 26 °C.

Over the complete event window, V1 remains above 26 °C for 66 h. The multi-zone variants remain above the threshold for shorter durations, ranging from 41 h for V2 to 54 h for V5 and V6. The maximum weighted mean air temperature occurs on 4 August at 16:00 and ranges from 28.71 °C for V4 to 29.10 °C for V1. The minimum weighted mean air temperature occurs on 6 August at 05:00 and ranges from 24.50 °C for V2 to 25.63 °C for V1.

Figure 4.9 presents the zone-level air temperatures for selected multi-zone variants during the same overheating event window. The zones follow a similar diurnal pattern, with temperatures increasing during the daytime and decreasing during the night and early morning. However, the temperature levels and durations above the

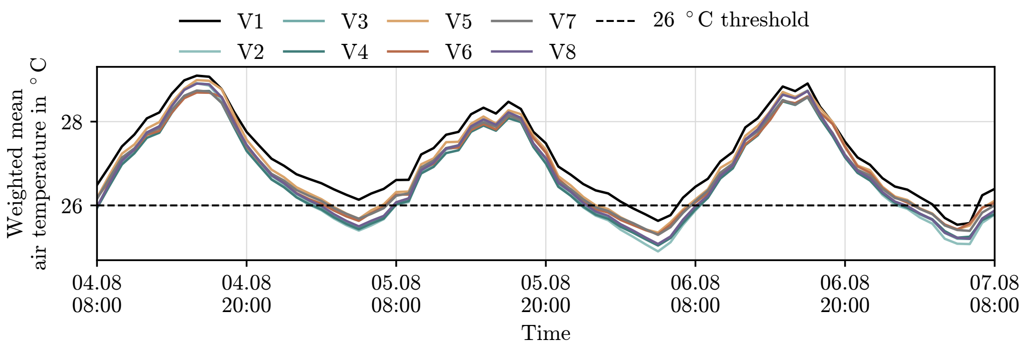


Fig. 4.8: Weighted mean air temperature during a selected overheating event for the zoning variants. The dashed horizontal line marks the overheating threshold of 26 °C.

26 °C threshold differ among the individual zones. The finer zoning variants show a wider spread of zone-level temperatures than the more aggregated variants.

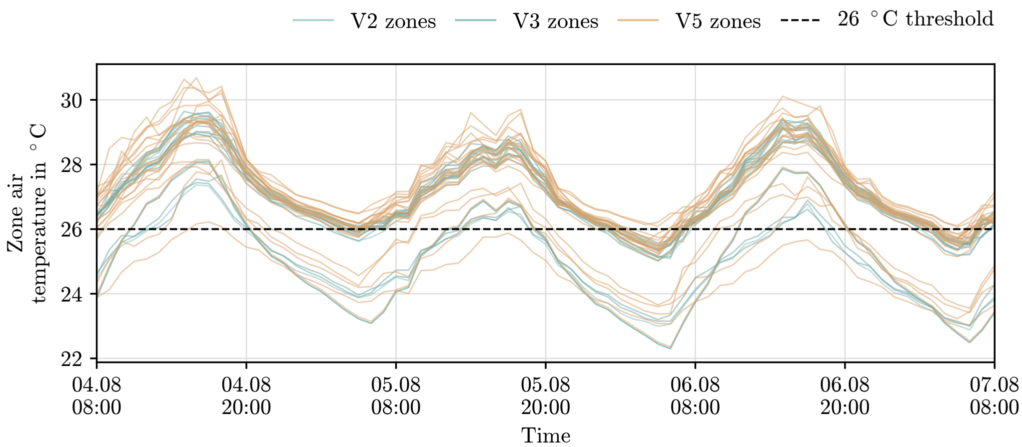


Fig. 4.9: Zone-level air temperatures during a selected overheating event for selected multi-zone variants. The dashed horizontal line marks the overheating threshold of 26 °C.

Supplementary results on the computational effort of the zoning variants are provided in Sec. C.1.

5 Discussion

This chapter interprets the results of the geometric validation and the zoning-variant sensitivity analysis. The discussion first evaluates the reliability of the Multizone Assignment Algorithm at the dwelling–stairwell abstraction level. It then examines how algorithm-related zoning deviations influence annual heating demand, peak heating load, and overheating hours before synthesising the contribution, limitations, future research directions, and impact of the thesis.

5.1 Geometric and topological reliability of the Multizone Assignment Algorithm

This section interprets the validation results with respect to the geometric and topological reliability of the generated dwelling–stairwell zoning.

5.1.1 Validation performance of the generated zoning output

The validation results indicate that the Multizone Assignment Algorithm can reproduce the dominant dwelling–stairwell zoning structure for a large proportion of the investigated multi-family residential floor plans. As shown in Fig. 4.1 and Tab. 4.1, more than four-fifths of the cases in both validation datasets achieve either a good or moderate level of geometric and topological agreement. This result provides evidence that the algorithm can generally approximate the intended dwelling–stairwell structure at the selected abstraction level.

The comparable outcome pattern across the Swiss and German datasets strengthens the interpretation that the results do not arise solely from one specific dataset format or extraction procedure. This point is important because the two datasets differ substantially in their origins, representations, and ground-truth extraction workflows. The Modified Swiss Dwellings cases provide a larger and more structured Swiss reference source. In contrast, the German real estate cases provide a smaller but more heterogeneous collection of German floor plan documents. Since both datasets show a similar dominance of good and moderate outcomes, the Multizone Assignment Algorithm appears to perform consistently across these two forms of floor plan evidence.

This result suggests that the algorithm can generally approximate dwelling and stairwell zoning at the intended abstraction level. The combined datasets also provide a stronger basis for interpreting the validation results within a Swiss–German residential building context than either dataset would provide alone. At the same time, the remaining deviation-related cases show that the proportion of good and moderate outcomes alone cannot describe the overall performance of the algorithm. The diagnostic categories and representative overlays therefore help explain where and why the algorithm-generated zoning differs from the reference layouts. The following subsections discuss these deviations.

5.1.2 Interpretation of good and moderate validation cases

In the representative good-match cases in Fig. 4.2 and Fig. 4.4, the algorithm-generated dwelling zones generally follow the reference dwelling extents, and the algorithm positions the stairwell close to the reference stairwell. These cases indicate that the Multizone Assignment Algorithm can preserve the principal spatial structure required for dwelling-level thermal zoning, including the approximate number and locations of dwelling zones and their relationship to the stairwell.

This outcome corresponds to the design of the Multizone Assignment Algorithm. The algorithm groups discretised footprint elements into dwelling zones using the inferred stairwell, access relations, façade contact, and area-balance constraints. This design supports good matches when the reference layout has a clear dwelling–stairwell organisation and when the estimated subdivision of the floor area corresponds closely to the reference dwelling structure (Waiz et al., 2025).

The good-match cases therefore represent successful reconstruction at the intended dwelling–stairwell abstraction level. Small local deviations along internal boundaries or stairwell edges remain acceptable when the algorithm preserves the dominant dwelling–stairwell organisation because the validation evaluates dwelling- and stairwell-level zoning rather than exact room-level geometry.

The moderate validation cases represent an intermediate level of agreement. In these cases, the algorithm usually captures the general dwelling–stairwell structure but produces weaker geometric overlap, local boundary deviations, or less accurate zone-area distributions. These cases do not constitute complete failures. Instead, they show that the Multizone Assignment Algorithm can preserve the principal thermal zone organisation even when it reproduces individual zone boundaries with lower geometric precision.

5.1.3 Qualitative analysis of deviation-related cases

The deviation-related cases provide additional insight into the conditions under which the algorithm produces less reliable zoning. Although the overall validation performance is positive, the representative deviation-related cases in Fig. 4.3 and Fig. 4.5 indicate recurring rather than random deviation mechanisms. These deviations mainly involve structural mismatch, under-zoning, over-zoning, and zones with shifted positions or high area errors.

The observed deviation-related cases can be interpreted through input-related mechanisms and implementation-related mechanism. The first mechanism concerns the limited input information and the structural assumptions imposed during preprocessing, particularly the inferred location of the stairwell. Since the actual internal layout and access organisation are unavailable, the workflow estimates the stairwell position from external and auxiliary information. The algorithm subsequently uses this position as a starting condition for generating the dwelling zones (Waiz et al., 2025). A difference between the inferred and reference stairwell organisation can therefore alter the resulting dwelling–stairwell topology. These cases are more critical than local boundary deviations because they affect the spatial relationships between the dwellings and the stairwell.

The second mechanism concerns the estimated dwelling subdivision and the area-allocation constraints. Since the algorithm derives the expected dwelling subdivision from estimated auxiliary information, a difference between this estimate and the reference layout can produce too few or too many dwelling zones and thereby lead to under-zoned or over-zoned cases. Such estimation uncertainty is common in urban-scale residential modelling, where modellers often estimate household areas and dwelling-related attributes from census, geometric, and statistical information rather than observe the internal layout directly (Köhler et al., 2021).

Even when the algorithm generates approximately the correct number of zones, it may still deviate from the reference layout because its objective favours compact and comparatively area-balanced dwelling zones based on the specified average dwelling area and size constraints (Waiz et al., 2025). However, dwelling areas within real multi-dwelling floor plans can vary substantially within the same building (Massafra et al., 2025). The algorithm-generated subdivision may therefore preserve the correct number of dwellings while differing from the actual dwelling-size distribution, which can lead to shifted internal boundaries or high area errors.

The third mechanism can contribute to local geometric deviations through the discretisation and post-processing of the generated zones. Binary Space Partitioning divides the footprint into discrete leaves that form the basic spatial units of the final zone assignment. Consequently, generated dwelling boundaries follow combinations of these partition elements rather than arbitrary architectural boundaries. The subsequent post-processing improves connectivity and removes local geometric irregularities, but it can also locally reassign or regularise partition elements (Waiz et al., 2025). These operations can therefore produce small boundary offsets, area differences, or shifted zones, particularly in irregular footprints and around the stairwell, even when the principal dwelling–stairwell organisation remains correct.

the deviation-related cases indicate that three input-related factors primarily influence the reliability of the Multizone Assignment Algorithm: the external footprint geometry, the inferred stairwell representation, and the estimated dwelling subdivision. The interaction of these inputs with the discretisation, area-allocation, and post-processing procedures can additionally contribute to local differences in zone boundaries, areas, and positions. These cases therefore do more than identify less reliable outputs. They reveal the principal conditions under which users should interpret algorithm-generated dwelling–stairwell zoning with caution.

5.1.4 Reliability of the Multizone Assignment Algorithm for dwelling–stairwell zoning

Taken together, the validation results indicate that the Multizone Assignment Algorithm generally reproduces dwelling- and stairwell-based zoning reliably at the intended abstraction level. In both validation datasets, the diagnostic classification assigns most cases to the good geometric match category.

However, this reliability remains conditional rather than universal. The good and moderate cases show that the method performs well when the external footprint, stairwell representation, and estimated dwelling subdivision correspond closely to the reference floor plan structure. In such cases, the remaining deviations mainly consist of local boundary differences and do not fundamentally alter the dwelling–stairwell organisation. By contrast, the deviation-related cases indicate that the output becomes less reliable when these assumptions do not adequately represent the reference layout, particularly when the predefined stairwell location or estimated dwelling subdivision differs from the reference layout.

The results therefore support the use of the Multizone Assignment Algorithm as an automated approximation of dwelling–stairwell zoning for many multi-family residential buildings, but not as a method for reconstructing detailed internal floor-plan geometry. This interpretation corresponds to the role of thermal zoning as a modelling abstraction, in which zones represent heat-balance and control units rather than exact architectural rooms or wall layouts (Johari et al., 2022; Shin and Haberl, 2019). The algorithm instead generates dwelling- and stairwell-based thermal zones from limited geometric input for simulation.

5.2 Sensitivity of simulation outputs to zoning deviations

The validation results identify the main geometric and topological conditions under which algorithm-related zoning deviations occur. The sensitivity analysis examines the influence of these deviation mechanisms on the simulation through controlled zoning variants. The discussion refers to the consolidated zoning-variant comparison in Fig. 4.6 and the median performance values in Tab. 4.2.

5.2.1 Annual heating demand

The results in Fig. 4.6(a) and Tab. 4.2, the transition from the fully aggregated V1 model to explicit multi-zone representations produces the dominant zoning-related effect on annual heating demand. V1 consistently produces the lowest median annual heating demand. However, this lower demand does not demonstrate higher accuracy or better model performance. Its interpretation must account for the TEASER–AixLib model formulation: the multi-zone variants include explicit internal separating surfaces and interzonal heat-transfer connections, whereas the single-zone model omits or aggregates these effects (Groesdonk et al., 2023).

Aggregation also smooths local effects such as dwelling-level temperature differences, internal gains, heat losses, setpoint differences, and interzonal heat exchange within one heat-balance unit. The uncertainty bands support this interpretation. The separation between V1 and the multi-zone variants is clearer than most differences within the multi-zone group, where the 5th–95th percentile ranges often overlap. Annual heating demand therefore responds more strongly to the transition from aggregated to multi-zone modelling than to most layout differences among the multi-zone variants (Johari et al., 2022; Kühn et al., 2024).

Within the multi-zone group, zoning granularity does not produce a linear effect. The difference between the coarser V2 variant and V3 remains small across the weather and construction states and ranges from approximately $-5.1\,\%$ to $+1.8\,\%$ relative to V3. In contrast, the finer V5 variant produces lower median annual heating demand than V3 in all cases, with reductions of approximately $5\,\%$ in the standard state and up to approximately 13 to $17\,\%$ in the retrofit state. Increased zoning granularity therefore does not automatically increase annual heating demand. Its effect depends on how the additional zones redistribute floor area, façade exposure, internal gains, stairwell adjacency, and interzonal heat exchange. The V3–V5 comparison consequently represents a configuration-dependent over-zoning effect rather than a general rule that additional zones always increase or decrease heating demand.

The layout comparisons indicate that shifted zones, high area errors, and moderate stairwell-position changes exert a smaller influence on annual heating demand. The difference between V3 and V4 remains below approximately $3\,\%$, while the difference between V5 and V6 remains at approximately 2.5 % or less. These small differences and the partially overlapping uncertainty ranges suggest that annual aggregation partly averages out layout and boundary-placement deviations when the overall zoning structure remains similar.

The stairwell-related variants indicate that the thermal representation of the stairwell becomes particularly relevant when its thermal state changes. V7, which omits a separate unheated stairwell zone, differs from V3 by only approximately 1 to 2 %. By contrast, V8 heats the stairwell and increases annual heating demand, with relative differences of approximately 4 to 5 % in the retrofit state. The stairwell therefore exerts less influence when the model either retains it as an unheated zone or omits it as a separate zone, but its influence increases when the model includes it in the heated volume.

Annual aggregation explains this behaviour because annual heating demand represents an energy-balance indicator over the full simulation year. When all variants retain the same external geometry, construction state, weather case, and overall residential use, the annual calculation partly averages local zoning differences over time and across the building. The annual heating-demand results therefore show that algorithm-related zoning deviations become most relevant when they alter the effective thermal structure of the model, such as the distinction between single-zone and multi-zone representation or the thermal treatment of the stairwell.

Within the multi-zone group, reduced granularity produces only a small effect, increased subdivision produces a clearer but configuration-dependent change, and layout-related differences remain limited where the uncertainty ranges overlap. Retrofit reduces the absolute demand level but makes the remaining variability more important in relative terms because the demand decreases more strongly than the uncertainty range.

5.2.2 Peak heating load

Peak heating load shows the same broad separation as annual heating demand: V1 remains below the explicit multi-zone variants across all investigated weather and construction states, as shown in Fig. 4.6(b) and Tab. 4.2. For peak heating load, however, this difference is particularly relevant because aggregation smooths short-term and zone-specific load peaks, while peak-load estimates inform heating-system sizing. The lower value produced by V1 should therefore not be interpreted as evidence of greater model accuracy.

The event-window analysis in Fig. 4.7 supports this interpretation from a temporal perspective. The peak occurs at approximately the same time across all variants, which indicates that the weather primarily determines the timing of the event. However, the magnitude of the heating load response differs among the zoning variants. The aggregated V1 model produces the smoothest and lowest load profile, while the multi-zone variants produce higher short-term peaks. Algorithm-related zoning differences therefore mainly affect the magnitude of the peak rather than the timing of the peak heating event.

Among variants with similar zoning resolution, the differences remain smaller, and the 5th–95th percentile uncertainty ranges overlap strongly. The coarser V2 variant remains close to V3, which suggests that a moderate reduction in zoning granularity has only a limited influence on peak heating load when the principal dwelling–stairwell structure remains comparable. By contrast, the finer V5 and V6 variants generally produce higher peak loads than V3 and V4, particularly under the cold-weather condition. Finer subdivision can therefore reveal local peak-load contributions more clearly because individual zones respond more independently to setpoints, heat losses, interzonal heat exchange, and stochastic internal gains.

The layout comparisons indicate that shifted zones, high area errors, and moderate stairwell-position changes have less influence on peak heating load than the broader change in zoning resolution. The differences between V3 and V4 and between V5 and V6 remain limited compared with the difference between V1 and the multi-zone variants. The small layout-related differences and overlapping uncertainty bands suggest that the representation of multiple thermal zones affects peak heating load more strongly than the exact placement of internal boundaries.

The stairwell-related variants show that the thermal representation of the stairwell can still affect peak-load estimation. V7 omits the separate unheated stairwell zone, while V8 includes the stairwell in the heated volume. Both changes alter the peak-load response relative to V3. The stairwell influences peak heating load because it can either function as an unheated buffer zone or contribute to the heated volume. The peak-load results therefore indicate that peak heating load responds more strongly to zoning aggregation, finer subdivision, and stairwell representation than to small layout shifts. However, the overlapping 5th–95th percentile ranges also indicate that some zoning-deviation effects have a magnitude comparable to the propagated input uncertainty.

5.2.3 Overheating hours

Figure 4.6(c) and Tab. 4.2 show that overheating hours respond differently to zoning deviations from the heating-related indicators. While the transition from V1 to the multi-zone variants primarily drives annual heating demand and peak heating load, the weather condition, construction state, and representation of local thermal zones exert a stronger influence on overheating hours. The warm-weather condition produces substantially more overheating hours than the cold-weather condition. The retrofit state also increases overheating hours, even though it reduces annual heating demand and peak heating load. This pattern indicates a target conflict: retrofit measures improve heating performance but can increase overheating risk when they reduce heat losses.

The aggregated V1 variant produces the highest number of overheating hours under Napoli conditions for both construction states and also produces the highest value in the Munich retrofit case. Aggregation therefore distorts the assessment of overheating differently from its effect on heating demand. V1 represents the building as one thermal zone and consequently cannot distinguish cooler and warmer internal areas. It averages local differences among dwellings, the stairwell, and zones with different exposure conditions into one temperature. V1 therefore does not provide a reliable representation of local overheating risk, even when it supplies a simple building-level indicator.

At the multi-zone level, overheating hours respond more strongly to spatial zoning effects than annual heating demand. Reduced granularity, finer subdivision, and alternative layouts influence the distribution of internal gains, façade exposure, heat storage, and interzonal heat exchange among the zones. Layout-related differences such as V3–V4 and V5–V6 therefore matter more for overheating than for annual heating demand because overheating is a threshold-based indicator that depends on local temperature exceedance rather than only on the total building heat balance.

The zone-level event window in Fig. 4.9 supports this interpretation. The selected variants follow the same weather-driven daily pattern, but their individual zone temperatures differ in peak magnitude and duration above the 26 °C threshold.

Compared with the building-level weighted mean temperatures in Fig. 4.8, individual zones in the finer zoning variants show stronger local overheating. Finer subdivision therefore does more than increase the number of thermal zones. It reveals local overheating behaviour that an aggregated representation can conceal.

The 5th–95th percentile uncertainty ranges provide important context for interpreting these differences. Under the cold-weather condition, the absolute number of overheating hours remains low, and some differences among the variants are small relative to the uncertainty range. Under the warm-weather condition, particularly in the retrofit state, overheating occurs much more frequently, which increases the relevance of zoning-related differences to the interpretation of thermal risk. Algorithm-related zoning deviations therefore become most important for overheating when the building already operates close to or frequently exceeds the overheating threshold.

The interpretation of the finer V5 and V6 variants must also account for the occupancy-profile assignment. These variants retain the household profiles in a larger number of separate dwelling zones, which redistributes the internal gains spatially and can thereby affect the duration of threshold exceedance.

The overheating-hour results show that algorithm-related zoning deviations matter more for comfort-related threshold indicators than for annual heating demand. Aggregation conceals local thermal differences, while finer subdivision and alternative layouts change the duration and magnitude of temperature-threshold exceedance. Overheating assessment therefore requires greater attention to zoning structure than aggregated heating-energy indicators.

Overall, the sensitivity analysis shows that algorithm-related zoning deviations affect simulation outputs, but their importance depends on both the type of deviation and the evaluated performance indicator. Aggregation and strong reductions in zoning detail smooth zone-specific temperature differences and heating load peaks, which leads to lower annual heating demand and peak heating load than explicit multi-zone representations. Changes in stairwell representation also matter because the stairwell can either function as an unheated buffer zone or contribute to the heated volume, thereby affecting heat losses and heating demand. By contrast, smaller layout shifts and boundary-placement differences exert a more limited influence on annual heating demand, particularly when the uncertainty ranges overlap. For overheating assessment, local zoning differences become more important because threshold exceedance depends on the temperature behaviour of individual zones rather than only on the building-level average. Zoning deviations therefore have the greatest simulation relevance when they alter heat-loss pathways, heated volume, interzonal heat exchange, short-term peak-load behaviour, or local temperature-threshold exceedance.

5.3 Contribution to automated thermal zoning evaluation

This thesis contributes a combined geometric and simulation-oriented evaluation of the Multizone Assignment Algorithm. It treats the algorithm-generated dwelling–stairwell zones not only as spatial outputs but also as modelling assumptions that influence the thermal representation of multi-family residential buildings. The study provides systematic evidence on the reliability of the generated zoning, identifies the main mechanisms through which it deviates from reference structures, and evaluates whether representative zoning differences influence selected building energy simulation outputs.

The first contribution is the independent, multi-case geometric validation of the Multizone Assignment Algorithm. Automated thermal zoning methods have previously been developed to infer simulation zones from limited building geometry, particularly through generic subdivisions or simplified spatial representations (Dogan et al., 2016; Xiang et al., 2024). The Multizone Assignment Algorithm addresses this limitation by generating dwelling–stairwell zoning from limited geometric and auxiliary input (Waiz et al., 2025). However, the original algorithm study demonstrated the workflow using selected building layouts and identified more extensive validation against real internal structures as an important research need. This left a lack of systematic evidence on the geometric and topological reliability of Multizone Assignment Algorithm-generated dwelling–stairwell zoning. The present study addresses this gap by comparing the generated zoning with independent reference structures derived from the Modified Swiss Dwellings dataset and German real estate floor plans. As shown in Tab. 4.1, good and moderate outcomes together account for more than four-fifths of both validation datasets. This provides broader empirical evidence that the algorithm can reproduce the principal dwelling–stairwell structure for many of the investigated small- to medium-sized multi-family residential buildings.

A further methodological contribution is the development of a validation basis for dwelling–stairwell zoning. Floor plans contain geometric, semantic, and topological information, but available datasets do not generally provide ready-made reference thermal zones at the dwelling–stairwell abstraction level. Moreover, floor plan information is commonly available through heterogeneous raster, vector, or graph-based representations, and differences in notation and document quality complicate their direct use (Kalervo et al., 2019; Pizarro et al., 2023; Wu et al., 2019). This thesis demonstrates how structured floor plan data and heterogeneous real estate floor plan documents can be converted into a common reference representation containing dwelling and stairwell geometries, labels, locations, and adjacency relations. This ground-truth construction workflow enabled the independent validation of the Multizone Assignment Algorithm and provides a potentially transferable basis for evaluating comparable automated dwelling–stairwell zoning methods.

The validation contribution also extends beyond reporting an overall level of geometric agreement. The diagnostic framework distinguishes structural mismatch, under-zoning, over-zoning, topology differences, and deviations in zone area or position. It therefore identifies not only whether the generated zoning agrees with the reference structure, but also how the principal deviations appear. The deviation-related cases indicate that the external footprint, inferred stairwell representation, and estimated dwelling subdivision are the main factors associated with reliable and less reliable zoning outputs in the investigated sample. This adds explanatory knowledge to the earlier demonstration of the algorithm by identifying the assumptions and deviation mechanisms that require particular attention when interpreting its output. It also reflects the wider challenge of estimating dwelling counts and dwelling areas when building-specific information is unavailable (Köhler et al., 2021).

The second contribution is the simulation-oriented interpretation of representative zoning deviations. Previous studies have established that thermal zoning influences building energy and comfort results (Johari et al., 2022; Shin and Haberl, 2019). However, this general knowledge does not explain how representative deviations associated with automated dwelling–stairwell zoning influence different simulation outputs. This leaves limited evidence on the simulation relevance of algorithm-related zoning deviations. The zoning-variant sensitivity analysis addresses this gap by examining reduced and increased zoning granularity, alternative internal boundary placement, omission of an explicit stairwell zone, and changes in stairwell thermal treatment under uncertain physical, operational, occupancy-related, and weather conditions.

The simulation-oriented contribution lies in demonstrating that the relevance of an automated zoning deviation depends on both the deviation mechanism and the evaluated performance indicator. Deviations that alter zoning granularity, heat-transfer pathways, heated volume, or local zone conditions can have clearer simulation consequences than small internal boundary shifts that preserve the principal dwelling–stairwell organisation. This extends existing knowledge beyond the general observation that thermal zoning affects simulation results by identifying which types of automated zoning deviation are most relevant to different outputs.

Changes in internal boundary placement generally have a smaller influence on building-level heating indicators, particularly where the uncertainty ranges overlap. However, these differences become more relevant to overheating because temperature-threshold exceedance depends on how façade exposure, internal gains, thermal capacity, and interzonal heat exchange are distributed among individual zones. The thermal treatment of the stairwell can also be more consequential than its geometric presence alone when it changes the heated volume and the stairwell's function as a thermal buffer. These findings refine the existing knowledge that thermal zoning matters by showing that its simulation relevance depends on the specific deviation mechanism and the evaluated performance indicator.

The combined evaluation contributes a two-stage logic for assessing automated thermal zoning. Geometric validation first determines whether and how the algorithm-generated zoning differs from the reference dwelling–stairwell structure. The zoning-variant sensitivity analysis then determines whether representative differences of this kind influence the selected simulation outputs under uncertain conditions. This distinction is important because a geometrically visible deviation does not automatically produce a substantial simulation effect, whereas deviations that alter zoning granularity, heat-transfer pathways, local thermal conditions, or the thermal treatment of shared spaces can produce clearer consequences. By connecting geometric reliability with simulation consequences, the thesis provides a more complete basis for assessing automated zoning than geometric agreement or simulation comparison alone.

Together, the geometric validation and zoning-variant sensitivity analysis address the research aim from complementary perspectives. The geometric validation evaluates the extent to which the Multizone Assignment Algorithm reproduces reference dwelling–stairwell structures, while the zoning-variant sensitivity analysis examines how representative zoning deviations influence selected simulation outputs depending on the deviation mechanism and the evaluated performance indicator. In combination, the two analyses establish both the reliability of the generated spatial representation and the simulation relevance of representative zoning deviations.

5.4 Limitations

Several limitations should be considered when interpreting the results. The first concerns the availability, scope, and regional coverage of the validation data. Publicly accessible floor plans for multi-family residential buildings rarely provide suitable building-level layouts. The validation data therefore cover only two national contexts within the DACH region: Switzerland, through the Modified Swiss Dwellings dataset, and Germany, through real estate floor plan documents. Austrian floor plan data are not included. The results should consequently be interpreted as evidence for a Swiss–German subset of the DACH region rather than as a complete validation across the entire DACH building stock. The sample primarily represents small- to medium-sized buildings and underrepresents larger, more complex buildings.

A second limitation concerns the construction of the German real estate reference data. The floor plans were obtained from heterogeneous PDF- and image-based sources and required manual digitisation, scaling, classification, and correction. Although this procedure made it possible to construct dwelling–stairwell reference geometries from otherwise unavailable data, it also introduced potential interpretation, vectorisation, and scaling uncertainties. The validation therefore compares the Multizone Assignment Algorithm with manually prepared reference zoning rather than automatically extracted reference data.

A related limitation concerns the absence of complete building-specific input data for the validation cases. The available sources did not consistently provide the original CityGML geometry, address and access information, construction year, or regional building-stock attributes required by the standard preprocessing workflow. The study therefore derived simplified external footprints, standardised their orientation, inserted them into an existing CityGML template, and prepared the required auxiliary inputs separately. These steps enabled consistent processing, but they do not reproduce the complete preprocessing process used in an operational urban-scale application. The validation therefore mainly assesses the automatic zoning step under prepared input conditions. In practical applications without reference floor plans, users also cannot directly confirm the actual number of dwellings, stairwell position, or internal access topology.

The use of controlled zoning variants limits the scope of the sensitivity analysis. These variants represent algorithm-related deviation mechanisms such as aggregation, reduced and increased zoning granularity, alternative internal boundary placement, omission of the stairwell, and changed stairwell thermal treatment. However, they do not reproduce each observed validation case directly. The sensitivity results should therefore be interpreted as an analysis of representative zoning-deviation mechanisms rather than as a direct simulation of every observed validation error.

Another limitation concerns the scope of the simulation outputs and input assumptions evaluated. The analysis focuses on annual heating demand, peak heating load, and overheating hours. These indicators are relevant to energy performance, system sizing, and thermal comfort, but they do not cover all possible consequences of zoning deviations. Cooling demand, emissions, detailed HVAC operation, room-level comfort metrics, and system-control effects were not evaluated. The results also remain conditional on the uncertain inputs, value ranges, weather cases, and construction states selected for this study.

A final limitation is that the simulation results were not validated against measured building-performance data. Without measured reference data, the analysis quantifies relative differences between zoning representations rather than absolute prediction errors. The results therefore show how algorithm-related zoning assumptions influence simulation outputs within the tested modelling framework, but they cannot determine which variant most closely represents the actual performance of a real building. The transferability of the findings to other residential building types, construction systems, climates, and modelling contexts remains an additional limitation.

5.5 Future work

Future research can extend the validation to a broader and more diverse residential building dataset. A larger reference sample would allow researchers to test whether the observed algorithm performance and deviation mechanisms remain consistent across different building typologies, access configurations, footprint shapes, and stairwell arrangements.

Researchers could also extend the validation framework to room-level reference data. Such an extension would support the evaluation of more detailed thermal zoning methods, layout-reconstruction workflows, and other zoning algorithms. This approach would require the extraction of room boundaries, internal doors, corridors, and adjacency relations from floor plans, but it would provide a richer basis for comparing different levels of zoning detail.

The more automated extraction of zoning reference data from floor plan sources represents another important research direction. Floor plan parsing, semantic segmentation, and vectorisation methods could detect dwelling areas, stairwells, circulation spaces, and relevant boundaries with less manual correction (Kalervo et al., 2019; Pizarro et al., 2023; Wu et al., 2019). These methods could support the construction of larger and more consistent ground-truth datasets for validating automated thermal zoning methods.

Building on this diagnostic basis, future work could use the deviation-related cases to refine the assumptions that guide automated zone generation and thermal enrichment. By analysing stairwell positions across a larger collection of floor plans, researchers could identify recurring relationships among footprint shape, entrance location, façade orientation, plot access, and stairwell placement. These patterns, together with richer contextual and thermal input data, could support more reliable stairwell placement and dwelling subdivision without altering the algorithm's fundamental zoning logic.

Future studies could extend the zoning-variant sensitivity analysis to additional outputs and validation settings. They should test whether the observed zoning effects remain consistent across other residential buildings, heating, ventilation, and air-conditioning assumptions, occupancy profiles, weather files, and future climate scenarios. Where measured data are available, future studies should compare simulated heating demand, peak loads, and indoor temperatures with monitored building performance. They could also expand the performance indicators to include cooling demand, emissions, system operation, and more detailed thermal comfort metrics.

5.6 Impact

The impact of this thesis lies in improving the reliability and interpretability of automated building performance simulation workflows. By validating algorithm-generated dwelling–stairwell zoning and analysing how zoning deviations influence simulation outputs under uncertain input conditions, the thesis provides evidence that can support better modelling decisions in urban building energy modelling.

The thermal performance of multi-family residential buildings directly affects thermal comfort, heating costs, and overheating risk. When automated zoning methods produce misleading thermal zone representations, simulations may underestimate or overestimate heating demand, peak heating loads, or overheating hours. By identifying when algorithm-generated zoning is reliable and when its deviations influence simulation outputs, this thesis supports more informed interpretation of simulation results for renovation planning, comfort assessment, and overheating-risk evaluation. This contribution is particularly relevant to occupants because inappropriate modelling assumptions can lead to retrofit decisions that reduce heating demand while unintentionally increasing overheating risk.

The thesis supports more robust residential building-stock analysis by showing that zoning-related differences remain visible under uncertain enrichment assumptions in urban building energy models. This knowledge improves the interpretation of simulation-based retrofit planning and decarbonisation strategies, particularly where modellers work with limited building information. The findings also demonstrate the importance of considering overheating alongside heating-demand reduction because energy-efficient retrofits can alter the thermal behaviour of residential buildings under warm conditions.

Municipalities, planners, housing companies, and energy consultants increasingly rely on automated modelling workflows because urban-scale datasets often lack detailed building information. The validation and zoning-variant sensitivity methods developed in this thesis help identify which geometric assumptions and zoning-deviation mechanisms require particular attention. These methods can support more targeted collection of zoning-relevant information, especially dwelling-size data and stairwell representations, and improve the interpretation of large-scale energy assessments. Economically, this prioritisation can support a more efficient allocation of modelling effort and renovation investment by directing resources towards the zoning-related information and modelling decisions that are most relevant to the evaluated simulation outputs.

Although computational effort was not a primary objective of this study, the supplementary results in Sec. C.1 provide relevant context for the scalability of the zoning representations. They show that additional zoning detail can substantially increase simulation runtime and that runtime is not determined by zone count alone. For large-scale applications, the required spatial detail and its simulation relevance should therefore be balanced against the associated computational effort.

The observed sensitivity of peak heating load to thermal zoning is also relevant to heat-supply system design. Since the zoning representation affects peak heating-load estimates, it may influence the design values used to size heat-generation units, thermal storage, and district-heating infrastructure. This connection is relevant because peak-load estimates provide important inputs for the design and sizing of heat-supply systems (Résimont et al., 2021). Thermal zoning assumptions should therefore be considered when building energy simulations are used to provide such design inputs.

Taken together, the thesis improves understanding of the Multizone Assignment Algorithm not only as an automated zoning method but also as part of a wider simulation workflow. It contributes to the broader goal of making urban building energy modelling more transparent, reliable, and useful for residential retrofit planning, climate adaptation, and evidence-based policy support.

6 Conclusion

This thesis investigated whether the Multizone Assignment Algorithm can generate simulation-relevant dwelling–stairwell thermal zoning for multi-family residential buildings from limited inputs. The study addressed a central challenge in urban building energy modelling: district- and city-scale datasets commonly describe the external form of buildings but provide little or no information about their internal dwelling structure, stairwells, or access relations (Wang et al., 2022). As a result, modellers often rely on strongly simplified thermal zoning assumptions, even though the selected zoning determines how a simulation represents heat balances, internal gains, temperature control, interzonal heat exchange, and local thermal conditions (Shin and Haberl, 2019).

The theoretical background showed that automated thermal zoning can help bridge the gap between limited urban data and the spatial information required for building energy simulation. Existing automated zoning methods demonstrate that simulation zones can be generated from incomplete geometric input, while the Multizone Assignment Algorithm specifically targets the dwelling–stairwell organisation of multi-family residential buildings. However, two questions remained unresolved. First, the reliability of the generated dwelling–stairwell structures had not been systematically tested against independent reference zoning. Second, existing knowledge that thermal zoning affects simulation results did not establish how representative deviations produced by an automated dwelling–stairwell zoning workflow influence different performance indicators.

To address these gaps, this thesis combined geometric validation with a zoning-variant sensitivity analysis. The validation constructed independent dwelling–stairwell reference zoning from two different types of floor plan evidence: structured Modified Swiss Dwellings data and heterogeneous German real estate floor plan documents. Both sources were converted into a common ground-truth representation containing dwelling and stairwell geometries, labels, positions, and adjacency relations. The resulting reference structures were compared with the Multizone Assignment Algorithm output using geometric and topological metrics and a hierarchical diagnostic classification.

The geometric validation addresses the first research gap concerning the lack of systematic evidence on the geometric and topological reliability of Multizone Assignment Algorithm-generated dwelling–stairwell zoning. Good and moderate outcomes together account for 81.5 % of the Modified Swiss Dwellings cases and 85.0 % of the German real estate cases. The similar outcome pattern across two data sources with different formats and extraction procedures indicates that the findings do not depend solely on one form of reference data. For many of the investigated small- to medium-sized multi-family residential buildings, the algorithm reproduces the principal number, position, and dwelling–stairwell arrangement of the reference zones at the intended abstraction level.

The validation also shows that this reliability is conditional rather than universal. The deviation-related cases reveal recurring mechanisms behind less reliable outputs. Structural mismatches and differences in the inferred stairwell organisation can alter the dwelling–stairwell topology, while uncertainty in the expected dwelling subdivision can produce under-zoning or over-zoning. Even when the zone count remains correct, area-balancing assumptions can shift internal boundaries or produce substantial differences in zone areas and positions. The external footprint, inferred stairwell representation, and estimated dwelling subdivision therefore emerge as the main zoning-relevant factors that require attention when interpreting an algorithm-generated layout.

A further methodological contribution to addressing this gap is the development of the validation basis itself. Existing floor plan sources do not generally provide ready-made thermal zones at the dwelling–stairwell level. This thesis demonstrates how structured and document-based floor plan information can be abstracted into a common geometric and topological reference representation. The workflow therefore provides a basis for evaluating comparable automated zoning approaches rather than limiting the contribution to the performance of the Multizone Assignment Algorithm alone.

The zoning-variant sensitivity analysis addresses the second research gap concerning the limited understanding of how representative algorithm-related zoning deviations influence building energy simulation outputs. It examines controlled representations of reduced and increased zoning granularity, alternative internal boundary placement, omission of an explicit stairwell zone, and changes in stairwell thermal treatment under varying construction, operational, occupancy-related, and weather conditions. The findings show that these zoning-deviation mechanisms are not equally relevant to all simulation outputs.

Moderate reductions in zoning granularity generally produce limited changes in annual heating demand and peak heating load when the principal dwelling–stairwell organisation remains comparable. Additional subdivision produces clearer but configuration-dependent effects and does not consistently increase or decrease the evaluated indicators. Internal boundary placement has a smaller influence on building-level heating indicators but becomes more relevant to overheating because temperature-threshold exceedance depends on local zone conditions. Stairwell representation is most consequential when its thermal treatment changes the heated volume or its function as an unheated buffer.

Accordingly, geometric agreement and simulation relevance should be treated as complementary evaluation criteria. Neither criterion alone is sufficient to determine whether automatically generated zoning is appropriate for a particular modelling purpose.

The central conclusion of this thesis is therefore that the Multizone Assignment Algorithm is useful as an automated dwelling–stairwell zoning approximation for scalable and early-stage energy simulation when detailed floor plans are unavailable, particularly when the alternative is a strongly aggregated single-zone model. However, it does not replace verified internal building information, and the usability of a deviating output depends on the deviation type and the required level of simulation detail. The method is most appropriate when the objective is to obtain a plausible, spatially differentiated representation for building-stock analysis, comparative scenario assessment, or early modelling stages. Applications that require verified dwelling boundaries, detailed local comfort assessment, or high-consequence design decisions continue to require more reliable internal information. This distinction helps determine when approximate zoning is sufficient and when additional internal building information is required.

Future work can build on this framework by extending the reference sample to larger buildings, multiple stairwells, more complex access systems, and additional regional building typologies. Direct comparisons between matched reference and algorithm-generated layouts could quantify the consequences of individual deviation-related cases more precisely, while automated floor plan parsing could support the construction of larger reference datasets. Output plausibility checks and confidence indicators could also help users identify uncertain zoning results when no reference floor plan is available, and comparisons with measured energy use and indoor-temperature data could connect zoning reliability with absolute simulation accuracy.

The knowledge developed in this thesis can therefore support better modelling decisions before more data become available. It helps identify which deviations may be acceptable for a selected building-level indicator, which assumptions require caution, and where further information about dwellings or stairwells is likely to provide the greatest value. This allows modelling effort to be directed towards the spatial information that matters most rather than towards uniformly increasing detail.

Automated zoning will not remove uncertainty from urban building energy modelling, but it can make the treatment of missing internal information more explicit, testable, and interpretable. By connecting limited urban geometry with independently evaluated dwelling–stairwell representations and by distinguishing geometric disagreement from simulation relevance, this thesis provides a step towards urban building models that are both scalable and more reflective of how residential buildings are internally organised. With broader validation, improved input inference, and transparent quality assessment, automated thermal zoning can become a valuable component of future building-stock simulation workflows for renovation planning, overheating assessment, climate adaptation, and evidence-based urban energy policy.

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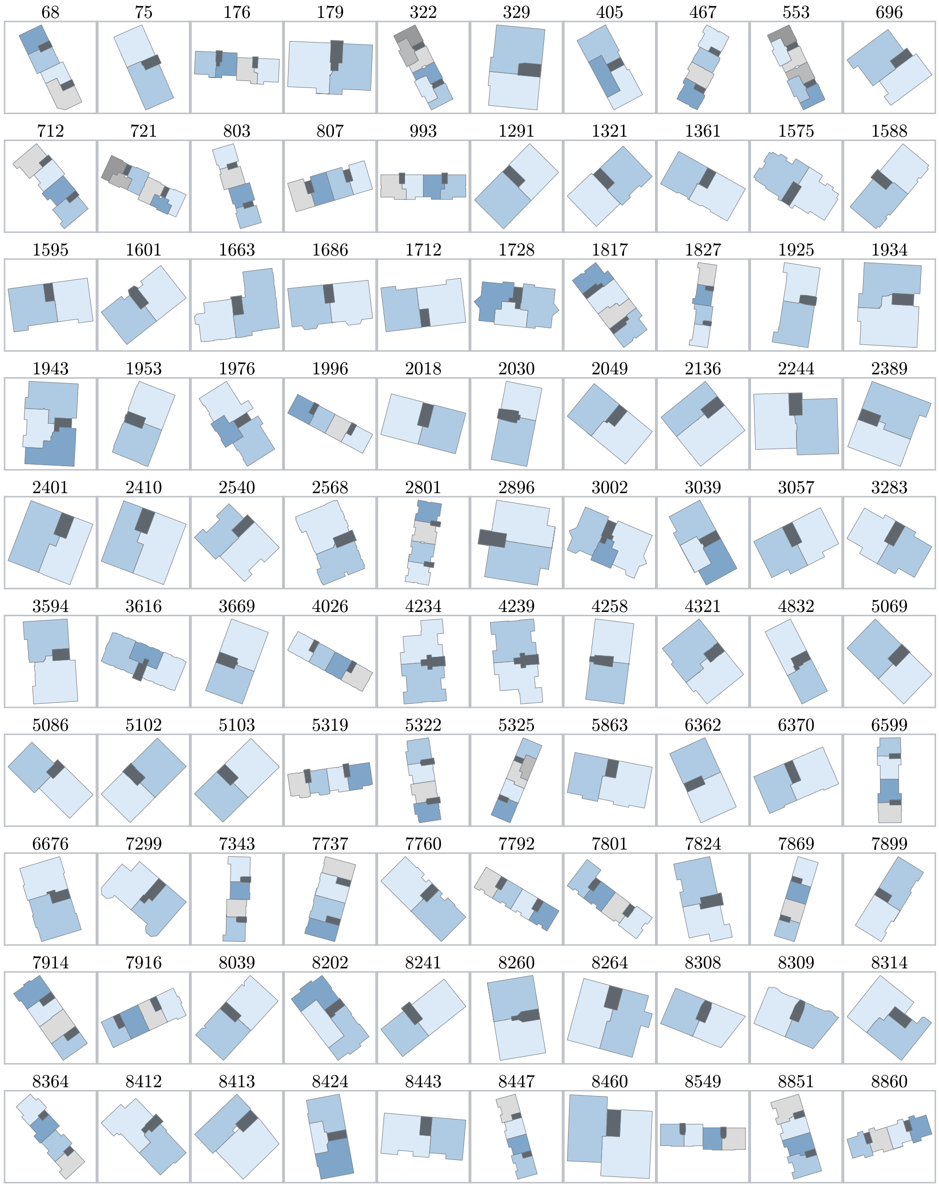
Appendix A

Reference ground-truth layouts

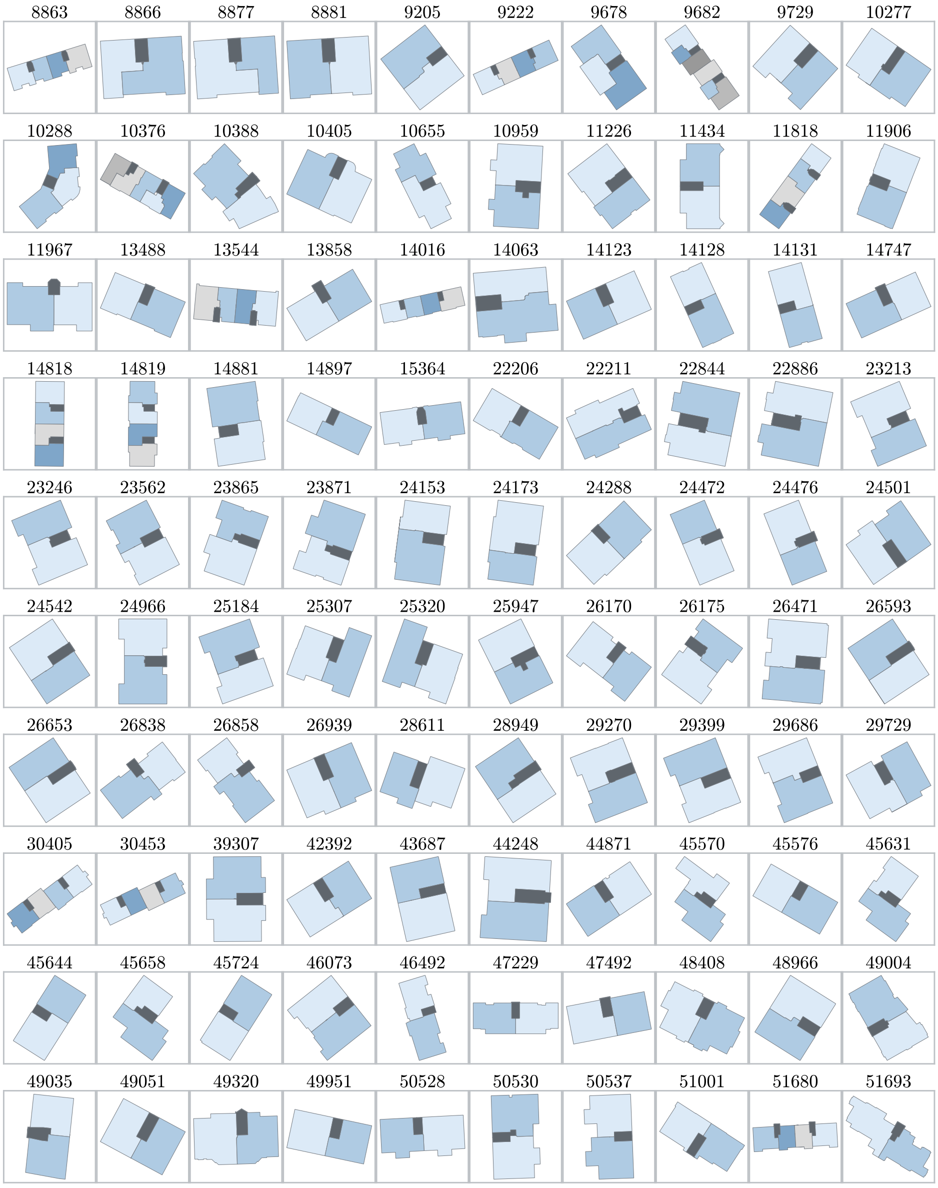
This appendix provides an overview of the reference ground-truth layouts selected for the geometric validation. The Modified Swiss Dwellings layouts show the dwelling and stairwell zoning representations that the extraction workflow produced after processing the room-level graph data and aggregating them to the selected validation abstraction level. Two panels present the 200 selected layouts to improve readability.

The German real estate layouts show the dwelling and stairwell zoning representations that the digitisation workflow derived from PDF- and image-based floor plan documents. The workflow manually digitised the layouts, scaled them to metric coordinates, and classified the resulting geometries into dwelling and stairwell zones. The figure includes the corresponding building and case identifiers to ensure traceability.



(a) First 100 selected layouts.

Fig. A.1: Selected Modified Swiss Dwellings ground-truth zoning layouts with corresponding building identifiers.



(b) Remaining 100 selected layouts.

Fig. A.1: Selected Modified Swiss Dwellings ground-truth zoning layouts with corresponding building identifiers (continued).

Fig. A.2: Selected German real estate ground-truth zoning layouts with corresponding case identifiers.

Appendix B

Auxiliary input estimation for Modified Swiss Dwellings cases

This appendix documents the auxiliary input estimation for the Modified Swiss Dwellings validation cases. The Modified Swiss Dwellings dataset does not provide the address, construction-year, or regional building-stock metadata that the standard Multizone Assignment Algorithm preprocessing workflow requires. This study therefore estimated the missing mean dwelling-area input from the external building footprint area.

This study used the estimation only to provide the auxiliary input required to run the Multizone Assignment Algorithm for the Modified Swiss Dwellings cases. The estimation did not form part of the reference ground-truth extraction, and this thesis does not interpret it as a representative dwelling-size model. The study did not directly use the known dwelling counts or measured dwelling areas of the validation cases as algorithm inputs because doing so would have introduced reference ground-truth information into the automated zoning process.

This study created a separate estimation dataset from Modified Swiss Dwellings buildings outside the validation sample. The dataset contained 400 buildings. For each building, the processing workflow derived the external building footprint area, dwelling areas, and stairwell area from the processed ground-truth geometries. The estimation models used the external building footprint area A_{fp} as the predictor and the mean dwelling area as the target variable.

The model-selection process compared several candidate models for estimating mean dwelling area from external building footprint area. These models included a constant baseline model, a simple linear regression model, and quantile-based piecewise linear regression models with different numbers of footprint-area regimes. The evaluation used a 70/30 train–test split of the estimation dataset and applied the mean absolute error as the primary model-selection metric.

Figure B.1 shows the mean absolute errors of the candidate estimation models. The model-selection process used this comparison to identify the model for the Modified Swiss Dwellings input-preparation workflow.

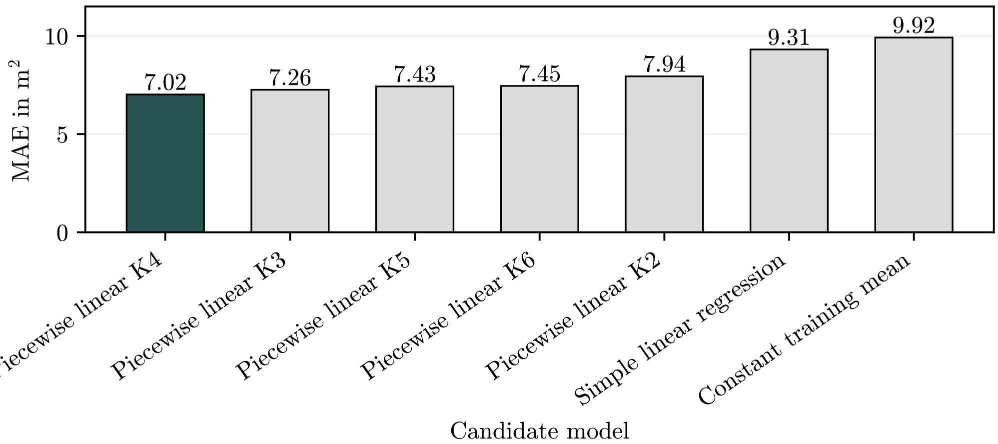


Fig. B.1: Comparison of candidate models for estimating mean dwelling area from external building footprint area.

The comparison showed that the quantile-based piecewise linear models achieved lower mean absolute errors than the constant baseline and simple linear regression models. Among the tested candidates, the four-regime piecewise linear model produced the lowest mean absolute error. The model-selection process therefore selected this model for the final implementation.

The selected model divides the footprint-area range into four regimes using quantile-based breakpoints. Within each regime, the model fits a separate linear relationship between external building footprint area and mean dwelling area. After model selection, the fitting procedure refitted the four-regime model using the complete estimation dataset to obtain the final parameters for the Multizone Assignment Algorithm input preparation.

The piecewise linear model has the general form

$$\hat{A}_{\mathrm{dw},r} = \alpha_r + \beta_r A_{\mathrm{fp}} \quad \text{for} \quad A_{\mathrm{fp}} \in I_r, \tag{B.1}$$

where $\hat{A}_{\mathrm{dw},r}$ denotes the estimated mean dwelling area in regime r, A_{fp} denotes the external building footprint area, I_r denotes the footprint-area interval associated with regime r, and α_r and β_r denote the fitted intercept and slope for that regime.

Figure B.2 shows the final fitted piecewise linear dwelling-area estimation model.

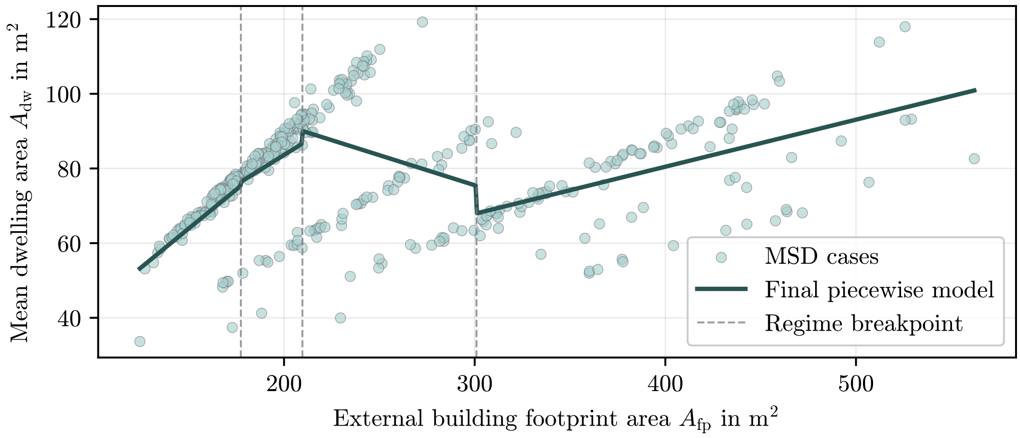


Fig. B.2: Final four-regime piecewise linear model for estimating mean dwelling area from external building footprint area. Vertical dashed lines mark the footprint-area breakpoints.

Equation (B.2) defines the final refitted four-regime model. All area values have the unit m^2.

$$\widehat{A}_{\mathrm{dw}} = \begin{cases} 1.2430 + 0.417820 A_{\mathrm{fp}} & \text{for } A_{\mathrm{fp}} \leq 177.3575, \\ 20.0725 + 0.317681 A_{\mathrm{fp}} & \text{for } 177.3575 < A_{\mathrm{fp}} \leq 209.6500, \\ 123.7882 - 0.160915 A_{\mathrm{fp}} & \text{for } 209.6500 < A_{\mathrm{fp}} \leq 300.9475, \\ 30.0455 + 0.126055 A_{\mathrm{fp}} & \text{for } A_{\mathrm{fp}} > 300.9475. \end{cases} \tag{B.2}$$

Appendix C

Detailed zoning-variant sensitivity analysis results

This appendix presents the numerical values supporting the zoning-variant sensitivity analysis described in Sec. 3.2. Table C.1 presents the results for the standard construction state, while Tab. C.2 presents the corresponding results for the retrofit construction state.

Tab. C.1: Median key performance indicator values and 5th–95th percentile ranges for the zoning variants in the standard construction state.

Variant	Annual heating demand in MWh		Peak heating load in kW		Overheating hours in h	
	Napoli	**Munich**	**Napoli**	**Munich**	**Napoli**	**Munich**
V1	71.7 [56.8–80.6]	175.2 [160.0–185.1]	39.5 [36.4–40.6]	61.1 [58.3–66.0]	2034 [1904–2314]	16 [7–77]
V2	96.9 [73.4–117.9]	224.9 [205.4–249.4]	48.2 [43.4–52.3]	78.8 [73.4–88.1]	1867 [1732–2084]	23 [8–54]
V3	95.7 [72.5–111.1]	223.8 [205.4–251.9]	47.8 [43.8–51.0]	79.2 [71.0–88.3]	1882 [1760–2108]	16 [7–53]
V4	97.0 [72.9–112.4]	225.5 [207.7–254.1]	47.8 [44.1–51.1]	79.3 [70.2–89.1]	1875 [1754–2076]	10 [3–32]
V5	90.5 [72.2–104.5]	213.2 [193.2–232.7]	49.2 [44.2–51.1]	83.6 [78.2–92.1]	1989 [1893–2188]	24 [8–64]
V6	90.6 [73.2–105.2]	211.7 [190.6–232.1]	48.8 [43.6–50.4]	80.7 [75.1–90.1]	1963 [1885–2123]	11 [3–33]
V7	93.7 [71.1–112.0]	222.2 [206.4–255.3]	48.8 [44.9–51.7]	83.2 [73.7–93.2]	1923 [1796–2148]	10 [4–39]
V8	97.2 [74.1–112.3]	228.8 [210.1–256.6]	49.2 [45.5–52.3]	85.5 [77.1–94.5]	1882 [1761–2110]	16 [7–53]

Note: The first line of each cell shows the median, and the second line shows the 5th–95th percentile range in square brackets. Overheating hours represent the annual number of hours during which at least one thermal zone exceeds 26 °C.

Tab. C.2: Median key performance indicator values and 5th–95th percentile ranges for the zoning variants in the retrofit construction state.

Variant	Annual heating demand in MWh		Peak heating load in kW		Overheating hours in h	
	Napoli	**Munich**	**Napoli**	**Munich**	**Napoli**	**Munich**
V1	22.7 [20.7–27.7]	49.0 [45.4–58.4]	23.7 [23.0–25.3]	59.1 [52.9–60.8]	2624 [2417–2767]	330 [131–427]
V2	35.8 [31.8–53.8]	76.2 [63.2–89.9]	30.7 [29.1–38.1]	73.5 [63.8–77.4]	2313 [2167–2444]	211 [118–292]
V3	37.6 [31.8–51.1]	75.5 [62.6–88.5]	31.1 [28.8–37.4]	72.1 [63.7–76.5]	2332 [2182–2476]	213 [114–292]
V4	38.6 [32.1–52.5]	76.5 [63.9–89.5]	31.3 [28.6–37.9]	72.2 [63.6–76.9]	2276 [2134–2402]	163 [86–248]
V5	32.3 [26.7–43.7]	62.2 [56.8–77.2]	32.3 [28.8–37.5]	79.6 [73.5–83.6]	2439 [2272–2544]	242 [116–319]
V6	33.1 [26.3–43.7]	63.2 [57.5–76.2]	31.3 [27.2–36.8]	77.4 [70.0–82.2]	2378 [2234–2477]	203 [85–265]
V7	37.3 [31.5–50.8]	74.3 [61.6–88.1]	32.2 [29.2–39.8]	75.2 [65.5–80.1]	2346 [2195–2507]	161 [77–246]
V8	39.0 [33.4–52.5]	79.0 [66.1–92.2]	33.0 [30.7–39.3]	78.3 [69.4–82.5]	2329 [2181–2476]	210 [111–291]

Note: The first line of each cell shows the median, and the second line shows the 5th–95th percentile range in square brackets. Overheating hours represent the annual number of hours during which at least one thermal zone exceeds 26 °C.

C.1 Computational effort

As a supplementary assessment, computational effort was evaluated for all 60 samples per zoning variant. Table C.3 presents the median simulation runtime for each zoning variant.

Tab. C.3: Median simulation runtime of the zoning variants based on 60 samples and 240 completed runs per variant.

Variant	Number of zones	Median simulation runtime in min
V1	1	0.59
V2	6	3.48
V3	11	6.97
V4	11	6.80
V5	21	15.90
V6	21	14.11
V7	10	2.20
V8	11	3.44

The median simulation runtime varied substantially among the zoning variants. V1 had the lowest median runtime at 0.59 min, while V5 and V6 had the highest median runtimes at 15.90 min and 14.11 min, respectively. The runtime generally increased for the more detailed multi-zone representations, but the variation was not determined by zone count alone. For example, V3, V4, and V8 each contained 11 zones but had median runtimes of 6.97 min, 6.80 min, and 3.44 min, respectively. The observed differences therefore also reflect variation in the resulting zoning and simulation-model configurations. However, the supplementary assessment did not separately examine which model characteristics caused these runtime differences.